

hard to compute and cumbersome to analyze. Usually, as we have seen in Section 17.C and in Appendix A of this chapter, it is possible to work with more aggregated, reduced systems. In fact, the general point duly made, it is worthwhile to observe that this is so even under the assumptions of Proposition 17.BB.2.⁸⁷ We elaborate briefly on this.

We can prove Proposition 17.BB.2 by setting up a two-player game instead of an $I + J + 1$ one.⁸⁸ The first player is an aggregate consumer-firm that has $\sum_i \hat{X}_i - \{\sum_i \omega_i\} - \sum_j \hat{Y}_j$ as its strategy set; the second is, as before, a market agent having Δ as its strategy set. Given $p \in \Delta$, the first agent responds with the set of vectors z expressible as $z = \sum_i x_i - \sum_i \omega_i - \sum_j y_j$, where y_j is profit maximizing in $\hat{Y}_j$ for every j , and $x_i \in \hat{X}_i$ is such that (1) $p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j$ and (2) $x_i \succeq_i x'_i$ whenever $p \cdot x'_i < p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j$. As before, the market agent responds with the set of $q \in \Delta$ that maximize $z \cdot q$ on Δ . Once this two-person game has been set up, the proof proceeds as for Proposition 17.BB.2. You should check this in Exercise 17.BB.9.

If for any $p \in \Delta$ the preference-maximizing choices of consumers, $x_i(p)$, and the profit-maximizing choices of firms, $y_j(p)$, were single valued, we could go one step further and consider a game with a single player (the market agent). Given p , we would then let the best response of the market agent be the set of price vectors $q \in \Delta$ that maximizes $[\sum_i x_i(p) - \sum_i \omega_i - \sum_j y_j(p)] \cdot q$ on Δ . In essence, this is what we did in the proof of Proposition 17.C.1.

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87. But it is not so for the generalizations described in the previous paragraph.

88. This was the approach taken in Debreu (1959).

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EXERCISES

17.B.1^A Show that for a pure exchange economy with $J = 1$ and $Y_1 = -\mathbb{R}_+^L$, “ $y_1^* \leq 0$, $p \cdot y_1^* = 0$, and $p \geq 0$ ” if and only if “ $y_1^* \in Y_1$ and $p \cdot y_1^* \geq p \cdot y_1$ for all $y_1 \in Y_1$.”

17.B.2^B Prove property (v) of Proposition 17.B.2. The proof of Proposition 17.B.2 in the text contains a hint. Recall also the following technical fact: any bounded sequence in $\mathbb{R}^L$ has a convergent subsequence.

17.B.3^B Suppose that $z(\cdot)$ is an aggregate excess demand function satisfying conditions (i) to (v) of Proposition 17.B.2. Let $p^n \rightarrow p$ with some, but not all, of the components of p being zero.

(a) Show that as n becomes large, the maximal excess demand is always obtained for some commodity whose price goes to zero.

(b) Argue (if possible by example) that a commodity whose price goes to zero may actually remain in excess supply for all n . [Hint: Relative prices matter.]

17.B.4^B Suppose that there are J firms whose production sets $Y_1, \dots, Y_J \subset \mathbb{R}^L$ are closed, strictly convex, and bounded above. Suppose also that a strictly positive consumption bundle is producible using the initial endowments and the economy's aggregate production set $Y = \sum_j Y_j$ (i.e., there is an $\bar{x} \gg 0$ such that $\bar{x} \in \{\sum_i \omega_i\} + Y$). Show that the production inclusive aggregate excess demand function $\bar{z}(p)$ in (17.B.3) satisfies properties (i) to (v) of Proposition 17.B.2.

17.B.5^A Suppose that there are J firms. Each firm produces a single output under conditions of constant returns. The unit cost function of firm j is $c_j(p)$, which we assume to be differentiable. The consumption side of the economy is expressed by an aggregate excess demand function $z(p)$. Write down an equation system similar to (17.B.4)–(17.B.5) for the equilibria of this economy.

17.B.6^C [Rader (1972)] Suppose that there is a single production set Y and that Y is a closed, convex cone satisfying free disposal. Consider the following exchange equilibrium problem. Given prices $p = (p_1, \dots, p_L)$, every consumer i chooses a vector $v_i \in \mathbb{R}^L$ so as to maximize $\succeq_i$ on the set $\{x_i \in X_i: p \cdot v_i \leq p \cdot \omega_i, \text{ and } x_i = v_i + y \text{ for some } y \in Y\}$. The price vector p and the choices $v^* = (v_1^*, \dots, v_J^*)$ are in equilibrium if $\sum_i v_i^* = \sum_i \omega_i$. Show that, under the standard assumptions on preferences and consumption sets, the price vector and the individual consumptions constitute a Walrasian equilibrium for the economy with production. Interpret.

17.C.1^A Verify that the correspondence $f(\cdot)$ introduced in the proof of Proposition 17.C.1 is convex-valued.

17.C.2^C Show that a convex-valued correspondence $z(\cdot)$ defined on $\mathbb{R}_{++}^L$ and satisfying the conditions (i) to (v) listed below (parallel to the corresponding conditions in Proposition 17.C.1) admits a solution; that is, there is a p with $0 \in z(p)$.

- (i) $z(\cdot)$ is upper-hemicontinuous.
- (ii) $z(\cdot)$ is homogeneous of degree zero.
- (iii) For every p and $z \in z(p)$ we have $p \cdot z = 0$ (Walras' law).
- (iv) There is $s \in \mathbb{R}$ such that $z_\ell > -s$ for any $z \in z(p)$ and p .
- (v) If $p^n \rightarrow p \neq 0$, $z^n \in z(p^n)$ and $p_\ell = 0$ for some ℓ , then $\text{Max}\{z_1^n, \dots, z_L^n\} \rightarrow \infty$.

[Hint: If you try to replicate exactly the proof of Proposition 17.C.1 you will run into difficulties with the upper-hemicontinuity condition. A possible three-step approach goes as follows: (1) Show that for $\varepsilon > 0$ small enough the solutions must be contained in $\Delta_\varepsilon = \{p \in \Delta: p_\ell \geq \varepsilon \text{ for all } \ell\}$; (2) argue then that for $r > 0$ large enough, one has $z(p) \subset [-r, r]^L$ for every $p \in \Delta_\varepsilon$; finally, (3) carry out a fixed-point argument in the domain $\Delta_\varepsilon \times [-r, r]^L$. For an easier result, you could limit yourself to prove the convex-valued parallel to Proposition 17.C.2. The suggested domain for the fixed-point argument is then $\Delta \times [-r, r]^L$.

17.C.3^B Consider an exchange economy in which every consumer i has continuous, strongly monotone, strictly convex preferences, and $\omega_i \gg 0$. The peculiarity of the equilibrium problem to be considered is that the consumer will now pay a type of tax on his gross consumption; moreover, this tax can differ across commodities and consumers. We will also assume that total tax receipts are rebated equally across consumers and in a lump-sum fashion. Specifically, for every i there is a vector of given tax rates $t_i = (t_{1i}, \dots, t_{Li}) \geq 0$ and for every price vector $p \gg 0$ the budget set of consumer i is

$$B_i(p, w_i) = \left\{ x_i \in \mathbb{R}_+^L : \sum_{\ell} (1 + t_{\ell i}) p_{\ell} x_{\ell i} \leq w_i \right\}.$$

An *equilibrium with taxes* is then a price vector $p \gg 0$ and an allocation $(x_1^*, \dots, x_I^*)$ with $\sum x_i^* = \sum \omega_i$ such that every i maximizes preferences in $B_i(p, p \cdot \omega_i + (1/I)(\sum_{\ell i} t_{\ell i} p_{\ell} x_{\ell i}))$.

(a) Illustrate the notion of an equilibrium with taxes in an Edgeworth box. Verify that an equilibrium with taxes need not be a Pareto optimum.

(b) Apply Proposition 17.C.1 to show that an equilibrium with taxes exists.

(c) As formulated here, the taxes are on gross consumptions. If they were imposed instead on net consumptions, that is, on amounts purchased or sold, then (assuming the same rate for buying or selling) the budget set would be

$$B_i(p, T_i) = \left\{ x_i \in \mathbb{R}_+^L : p \cdot (x_i - \omega_i) + \sum_{\ell} t_{\ell} |p_{\ell} (x_{\ell i} - \omega_{\ell i})| \leq T_i \right\},$$

where the T_i are the lump-sum rebates. In what way does this budget set differ from that described previously for the case of taxes on gross consumptions? Represent graphically. Notice the kinks.

(d) Write down a budget set for the situation similar to (c) except that the tax rates for amounts bought or sold may be different.

(e) (More advanced) How would you approach the existence issue for the modification described in (c)?

17.C.4^A Consider a pure exchange economy. The only novelty is that a progressive tax system is instituted according to the following rule: individual wealth is no longer $p \cdot \omega_i$; instead, anyone with wealth above the mean of the population must contribute half of the excess over the mean into a fund, and those below the mean receive a contribution from the fund in proportion to their deficiency below the mean.

(a) For a two-consumer society with endowments $\omega_1 = (1, 2)$ and $\omega_2 = (2, 1)$, write the after-tax wealths of the two consumers as a function of prices.

(b) If the consumer preferences are continuous, strictly convex, and strongly monotone, will the excess demand functions satisfy the conditions required for existence in Proposition 17.C.1 given that wealth is being redistributed in this way?

17.C.5^B Consider a population of I consumers. Every consumer i has consumption set $\mathbb{R}_+^L$ and continuous, strictly convex preferences $\succsim_i$. Suppose, in addition, that every i has a household technology $Y_i \subset \mathbb{R}^L$ satisfying $0 \in Y_i$. We can then define the *induced preferences* $\succsim_i^*$ on $\mathbb{R}_+^L$ by $x_i \succsim_i^* x'_i$ if and only if for any $y'_i \in Y_i$ with $x'_i + y'_i \geq 0$ there is $y_i \in Y_i$ with $x_i + y_i \geq 0$ and $x_i + y_i \succsim_i x'_i + y'_i$ (i.e., whatever can be done from x'_i , something at least as good can be obtained from x_i).

(a) Show that induced preferences are rational, that is, complete and transitive.

(b) Show that if Y_i is convex then induced preferences $\succsim_i^*$ are convex.

(c) Suppose that goods are of two kinds: marketed goods and nonmarketed household goods. Initial preferences $\succsim_i$ care only about household goods, and initial endowments ω_i have nonzero entries only for marketed goods. Use the concept of induced preferences to set up the equilibrium problem as one that is formally a problem of pure exchange among marketed goods. Discuss.

17.C.6^B Let $L = 2$. Consider conditions (i), (iii), and (iv) of Proposition 17.B.2. Exhibit four examples such that in each of the examples only one condition fails and yet the system of equations $z(p) = 0$ has no solution. Why is condition (ii) not included in the list?

17.D.1^B Consider an exchange economy with two commodities and two consumers. Both consumers have homothetic preferences of the constant elasticity variety. Moreover, the elasticity of substitution is the same for both consumers and is small (i.e., goods are close to perfect complements). Specifically,

$$u_1(x_{11}, x_{21}) = (2x_{11}^\rho + x_{21}^\rho)^{1/\rho} \quad \text{and} \quad u_2(x_{12}, x_{22}) = (x_{12}^\rho + 2x_{22}^\rho)^{1/\rho},$$

and $\rho = -4$. The endowments are $\omega_1 = (1, 0)$ and $\omega_2 = (0, 1)$.

Compute the excess demand function of this economy and verify that there are multiple equilibria.

17.D.2^A Apply the implicit function theorem to show that if $f(v) = 0$ is a system of M equations in N unknowns and if at $\bar{v}$ we have $f(\bar{v}) = 0$ and $\text{rank } Df(\bar{v}) = M$, then in a neighborhood of $\bar{v}$ the solution set of $f(\cdot) = 0$ can be parameterized by means of $N - M$ parameters.

17.D.3^A Carry out explicitly the computations for Proposition 17.D.4.

17.D.4^C Consider a two-commodity, two-consumer exchange economy satisfying the appropriate differentiability conditions on utility and demand functions. There is a total endowment vector $\bar{\omega} \gg 0$. Show that for almost every $\omega_1 \ll \bar{\omega}$ the economy defined by the initial endowments ω_1 and $\omega_2 = \bar{\omega} - \omega_1$ has a finite number of equilibria. This differs from the situation in Proposition 17.D.2 in that total endowments are kept fixed. [Hint: You should use the properties of the Slutsky matrix.]

17.D.5^B Consider a two-commodity, two-consumer exchange economy satisfying the appropriate differentiability conditions on utility and demand functions. Set the equilibrium problem as an equation system in the consumption variables $x_1 \in \mathbb{R}_+^2$ and $x_2 \in \mathbb{R}_+^2$, the price variables $p \in \mathbb{R}_+^2$, and the reciprocals of the marginal utilities of wealth $\lambda_1 \in \mathbb{R}$ and $\lambda_2 \in \mathbb{R}$ (neglect the possibility of boundary equilibria). The parameters of the system are the initial endowments $(\omega_1, \omega_2) \in \mathbb{R}_+^4$. Prove without further aggregation that (after deleting one equation and one unknown) the system satisfies the full rank condition of the transversality theorem.

17.D.6^B The setup is identical to Exercise 17.D.5 except that an externality is allowed: The (differentiable) utility function of consumer 1 may depend on the consumption of consumer 2; that is, it has the form $u_1(x_1, x_2)$ where x_i is consumer i 's consumption bundle [but we still have $u_2(x_2)$]. Equilibrium is defined as usual, with the proviso that consumer 1 takes consumer 2's consumption as given. Show that, generically on initial endowments $(\omega_1, \omega_2) \in \mathbb{R}_+^4$, the number of equilibria is finite.

17.D.7^B Suppose the agents of an overall exchange economy are distributed across N islands with no communication among them. Each island economy has three equilibria.

(a) Argue that the number of equilibria in the overall economy is 3^N .

(b) Suppose now that the islands' economies are identical and that there is a possibility of communication across the islands: free and costless transportation of commodities. Show that then the number of equilibria is 3.

17.D.8^A Show by explicit computation that the index of the equilibrium of a one-consumer Cobb–Douglas pure exchange economy is $+1$.

17.E.1^A Derive expressions (17.E.1) and (17.E.2).

17.E.2^A Derive expression (17.E.3).

17.E.3^B Provide explicit utility functions rationalizing at a given price vector p the individual excess demands $z_i(p)$ and matrices of price effects $Dz_i(p)$ constructed in the proof of Proposition 17.E.2.

17.E.4^B Consider the two-commodity case. Give an example of a function $z(p)$ defined on $P_\varepsilon = \{(p_1, p_2) \gg 0 : \varepsilon < (p_1/p_2) < (1/\varepsilon)\}$, and with values in $\mathbb{R}^2$, that is continuous, is homogeneous of degree zero, satisfies Walras' law, and cannot be generated from a rational preference relation. Represent graphically the offer curve associated with this function. Note that it goes through the initial endowment point and compare with the construction used in Figure 17.E.2.

17.E.5^A Show that the choices represented in Figure 17.E.3 cannot be generated from consumers with endowment vectors bounded above by $(1, 1)$ and nonnegative consumption.

17.E.6^A Show that the excess demand function $z_i(p) = e^i - p_i p$, defined for $\|p\| = 1$, is proportionally one-to-one in the sense used in the general proof of Proposition 17.E.3 (at the end of Section 17.E).

17.E.7^B Show directly that the excess demand function $z_i(p) = e^i - p_i p$ used in the general proof of Proposition 17.E.3 satisfies the strong axiom of revealed preference.

17.F.1^C Show that expression (17.F.2) gives rise to a negative semidefinite matrix of price effects, $Dz(p)$, if initial endowments are proportional among themselves *or* if consumptions are proportional among themselves.

17.F.2^A Complete the requested verification of Example 17.F.1.

17.F.3^B There are four goods and two consumers. The endowments of the consumers are $\omega_1 = (\omega_{11}, \omega_{21}, 0, 0)$ and $\omega_2 = (\omega_{12}, \omega_{22}, 0, 0)$. Consumer 1 spends all his wealth on good 3 while consumer 2 does the same on good 4. Specify some values of ω_1 and ω_2 for which the corresponding excess demand of this economy does not satisfy the weak axiom of revealed preference.

17.F.4^A Suppose that there are L goods but that for every consumer there is a good such that at any price the consumer spends all his wealth on that good (perhaps goods are distinguished by their location). Show that the aggregate excess demand will satisfy the (weak) gross substitute property.

17.F.5^C Complete the missing steps of Example 17.F.2.

17.F.6^C Consider a two-consumption-good, two-factor model with constant returns and no joint production. In fact, suppose that the production functions for the two consumption goods are Cobb–Douglas. Consumers have holdings of factors and have preferences only for the two consumption goods. The economy is a closed economy (at equilibrium, consumption must equal production). Suppose that the two goods are normal and gross substitutes in the *demand function* of the consumers. Define an induced exchange economy for factors of production by assuming that at any vector of factor prices the two goods are priced at average cost and the final demand for them is met. Show that the resulting aggregate excess demand for factors of production has the gross substitute property and, consequently, that there is a unique equilibrium for the overall economy.

17.F.7^A Prove expression (17.F.3) for $L = 2$.

17.F.8^A Show that expression (17.F.3) implies that the set of solutions to $z(p) = 0$ is convex.

17.F.9^B Consider an economy with a single constant returns production set Y . Preferences are continuous, strictly convex, and strongly monotone. Suppose that the feasible consumptions $(x_1, \dots, x_I)$ are associated with a Walrasian equilibrium. Assume, moreover, that no trade is required to attain these consumptions if Y is freely available to all consumers; that is $x_i - \omega_i \in Y$ for all i . Show then that those are the only possible equilibrium consumptions.

17.F.10^A Show that expression (17.F.3) implies that $Dz(p)$ is negative semidefinite at an equilibrium p .

17.F.11^B Show that if $z(p) = 0$, $\text{rank } Dz(p) = L - 1$, and $Dz(p)$ is negative semidefinite, then, for any ℓ , the $(L - 1) \times (L - 1)$ matrix obtained from $Dz(p)$ by deleting the ℓ th row and column has a determinant of sign $(-1)^{L-1}$. [Hint: From Section M.D of the Mathematical Appendix you know that $\text{rank } Dz(p) = L - 1$ implies that the $(L - 1) \times (L - 1)$ matrix under study is nonsingular. Consider then $Dz(p) - \alpha I$.]

17.F.12^B Show that if $z(p) = 0$ and $Dz(p)$ has the gross substitute sign pattern, then the $(L - 1) \times (L - 1)$ matrix obtained from $Dz(p)$ by deleting the ℓ th row and column has a *negative dominant diagonal* (see Section M.D of the Mathematical Appendix for this concept) and is therefore negative definite.

17.F.13^A Provide the missing computation for Example 17.F.3.

17.F.14^B Consider a firm that produces good 1 out of goods $\ell = 2, \dots, L$ by means of a production function $f(v_2, \dots, v_L)$. Assume that $f(\cdot)$ is concave, increasing, and twice continuously differentiable. We say that ℓ and ℓ' are complements at the input combination $v = (v_2, \dots, v_L)$ if $\partial^2 f(v) / \partial v_\ell \partial v_{\ell'} > 0$.

(a) Verify that for the Cobb–Douglas production function $f(v_2, \dots, v_L) = v_2^{\alpha_2} \times \dots \times v_L^{\alpha_L}$, $\alpha_2 + \dots + \alpha_L \leq 1$, any two inputs are complements at any v .

(b) Suppose that $f(\cdot)$ is of the constant returns type. Show that at any v and for any ℓ there is an ℓ' that is a complement to ℓ at v .

(c) Suppose now that $f(\cdot)$ is strictly concave and that any two inputs are complements at any v . Let $v_\ell(p_1, \dots, p_L)$ be the input demand functions. Show that, for any ℓ , $\partial v_\ell / \partial p_1 > 0$, $\partial v_\ell / \partial p_\ell < 0$, and $\partial v_\ell / \partial p_{\ell'} < 0$ for $\ell' \neq \ell$.

(d) Discuss the implications of (a) to (c) for uniqueness theorems that rely on the gross substitute property.

17.F.15^B Consider a one-consumer economy with production and strictly convex preferences. There is a system of ad valorem taxes $t = (t_1, \dots, t_L)$ creating a wedge between consumer and producer prices; that is, $p_\ell = (1 + t_\ell)q_\ell$ where p_ℓ and q_ℓ are, respectively, the consumer and producer price for good ℓ . Tax receipts are turned back in lump-sum fashion. Write the definition of (distorted) equilibrium. Show that the equilibrium is unique if the production sector is of the Leontief type (a single primary factor, no joint production, constant returns) and all goods are normal in consumption. Can you argue by example the nondispensability of the last normality condition? If this is simpler, you can limit your discussion to the case of two commodities (one input and one output).

17.F.16^C Suppose that $g(p) = (g_1(p), \dots, g_N(p))$ is defined in the domain $[0, r]^N$ and that $g(0, \dots, 0) \gg (0, \dots, 0)$, $g(r, \dots, r) \ll (0, \dots, 0)$. Note that we do not assume Walras' law, homogeneity of degree zero, or, for that matter, continuity. The function $g(\cdot)$ could, for

example, be the system of excess demands corresponding to a subgroup of markets with the prices of commodities outside the group kept fixed.

(a) We say that $g(\cdot)$ satisfies the *strong gross substitute property* (SGS) if for some $\alpha > 0$ every coordinate of the function $\alpha g(p) + p$ is strictly increasing in p and $(\alpha g(p) + p) \in [0, r]^N$ for every $p \in [0, r]^N$. Show that if $g(p)$ has the SGS property then it also has the GS property.

(b) Show by example that the GS property does not imply the SGS property. Establish, however, that if $g(\cdot)$ is continuously differentiable and the GS property is satisfied then the SGS property holds.

From now on we assume that $g(\cdot)$ satisfies the SGS property.

(c) Show that there is an equilibrium, that is, a p with $g(p) = 0$. Illustrate graphically for the case $N = 1$. [Hint: Quote the Tarski fixed point theorem from Section M.I of the Mathematical Appendix, or, if you prefer, assume continuity and apply Brouwer's fixed point theorem.]

(d) Give an example for $N = 2$ where the equilibrium is not unique.

(e) Suppose that $g(p) = g(p') = 0$. Show that there must be an equilibrium p'' such that $p'' \geq p$ and $p'' \leq p'$. Similarly, there is an equilibrium p''' such that $p''' \leq p$ and $p''' \leq p'$. [Hint: Apply the argument in (c) to the domain $[\text{Max}\{p_1, p'_1\}, r] \times \cdots \times [\text{Max}\{p_N, p'_N\}, r]$.]

(f) Argue (you can assume continuity here) that the equilibrium set satisfies a strong and very special property, namely, that it has a maximal and a minimal equilibrium. That is, there are $p^{\max}$ and $p^{\min}$ such that $g(p^{\max}) = g(p^{\min}) = 0$ and $p^{\min} \leq p \leq p^{\max}$ whenever $g(p) = 0$.

(g) Assume now that $g(\cdot)$ is also differentiable. Suppose that we know that at equilibrium, that is, whenever $g(p) = 0$, the matrix $Dg(p)$ has a *negative dominant diagonal*; that is, $Dg(p)v \ll 0$ for a $v \gg 0$. Argue (perhaps nonrigorously) that the equilibrium must then be unique.

(h) Suppose that $g(\cdot)$ is the usual excess demand system for the first N goods of an economy with $N + 1$ goods in which the last price has been fixed to equal 1 and the overall $(N + 1)$ -good excess demand system satisfies the gross substitute property. Apply (g) to show that the equilibrium is unique.

17.F.17^A [Becker (1962), Grandmont (1992)] Suppose that $L = 2$ and you have a continuum of consumers. All consumers have the same initial endowments; they are not rational, however. Given a budget set, they choose at random from consumption bundles on the budget line using a uniform distribution among the nonnegative consumptions. Let $z(p)$ be the average excess demand (= expected value of a single consumer's choice). Show that $z(\cdot)$ can be generated from preference maximization of a Cobb–Douglas utility function (thus the economy admits a positive representative consumer in the sense of Section 4.D).

17.G.1^B Suppose that in an exchange economy (and with the normalization $p_L = 1$) we are given equilibrium prices $p(\hat{\omega}_1)$ as a differentiable function defined as an open domain of the endowments of the first $L - 1$ goods of the first consumer, $\hat{\omega}_1 = (\omega_{11}, \dots, \omega_{L-1,1})$. All the remaining endowments are kept fixed. Suppose that the demand function of the first consumer is strictly normal in the sense that $D_{w_1} x_1(p, w_1) \gg 0$ through the relevant domain of (p, w_1) . Show then that for any $\hat{\omega}_1$ and $\bar{p} = p(\hat{\omega}_1)$, we have $\text{rank } D_{\hat{\omega}_1} \hat{z}_1(\bar{p}; \hat{\omega}_1) = L - 1$ and $\text{rank } Dp(\hat{\omega}_1) = L - 1$, where $\hat{z}_1(p; \hat{\omega}_1)$ is the excess demand function of the first consumer for the first $L - 1$ goods.

17.G.2^B The setting is as in Exercise 17.G.1 or as in Proposition 17.G.2. Suppose that $\hat{z}(\bar{p}; \hat{\omega}_1) = 0$. Show that there are economies with $D_p \hat{z}(\bar{p}; \hat{\omega}_1)$ an $(L - 1) \times (L - 1)$ negative definite matrix but where $\partial p_1(\hat{\omega}_1) / \partial \omega_{11} > 0$. [Hint: Use Proposition 17.G.1 and the arguments employed in its proof.]

17.G.3^C The setting is as in Exercise 17.F.16, except that now we have two functions $g(p) \in \mathbb{R}^N$ and $\hat{g}(p) \in \mathbb{R}^N$. Each of these functions satisfies the conditions of Exercise 17.F.16 (in particular the SDS property). In addition, we assume that $\hat{g}(\cdot)$ is an upward shift of $g(\cdot)$; that is, $\hat{g}(p) \geq g(p)$ for every $p \in [0, r]^N$. Prove that if $(p^{\min}, p^{\max})$ and $(\hat{p}^{\min}, \hat{p}^{\max})$ are the minimal and maximal equilibrium price vectors (see Exercise 17.F.16) for $g(\cdot)$ and $\hat{g}(\cdot)$, respectively, then $\hat{p}^{\min} \geq p^{\min}$ and $\hat{p}^{\max} \geq p^{\max}$. [You can assume that $g(\cdot)$ and $\hat{g}(\cdot)$ are continuous; if this makes things simpler, assume also that both functions have a unique solution.] Represent graphically for the case $N = 1$.

17.H.1^C Suppose that the system of excess demand functions $z(p)$ satisfies the gross substitute property. Consider the tâtonnement price dynamics

$$\frac{dp_\ell}{dt} = z_\ell(p) \quad \text{for every } \ell. \quad (*)$$

For any price vector p let $\psi(p) = \text{Max} \{z_1(p)/p_1, \dots, z_L(p)/p_L\}$.

(a) Argue that if $p(t)$ is a solution for the above tâtonnement dynamics (i.e., $dp_\ell(t)/dt = z_\ell(p(t))$ for every ℓ and t) and $z(p(0)) \neq 0$ then $\psi(p(t))$ should be decreasing through time. [Hint: If $z_\ell(p(t))/p_\ell(t) = \psi(p(t))$ then $p_\ell(t)/p_{\ell'}(t)$ cannot decrease at t for any ℓ' . Hence, $z_\ell(p(t))$ cannot increase, whereas p_ℓ surely increases.]

(b) Argue that $p(t)$ converges to an equilibrium price as $t \rightarrow \infty$. [Hint: Recall that for the dynamics (*) Walras' law implies that $\sum_\ell p_\ell^2(t) = \text{constant}$.]

17.H.2^B There is an output good and a numeraire. The price of the output good is p . The data of our problem are given by two functions: The consumption side of the economy provides an excess demand function $z(p)$ for the output good, and the production side an increasing inverse output supply function $p(z)$. Both functions are differentiable. In addition, their graphs cross at $(1, 1)$, which is the equilibrium we will concentrate on in this exercise.

Given this setting we can define two one-variable dynamics:

- (i) In *Walras price dynamics* we assume that at p the price increases or decreases according to the sign of the difference between excess demand and (direct) supply at p .
- (ii) In *Marshall quantity dynamics* we assume that at z production increases or decreases according to the sign of the difference between the demand price (i.e., the inverse excess demand) and the supply price (i.e., $p(z)$) at z .

(a) Write the above formally and interpret economically.

(b) Suppose that the technology is nearly of the constant returns type. Show then that around the equilibrium $(1, 1)$ the system is always Walrasian stable but that Marshallian stability depends on the slope of the excess demand function (in what way?).

(c) Write general price and quantity dynamics where prices move à la Walras and quantities à la Marshall. Draw a (p, z) phase diagram and argue that in the typical case dynamic trajectories will spiral around the equilibrium.

(d) Go back to the technology specification of (b). Show that the system in (c) is locally stable if and only if the equilibrium is Marshallian stable.

(e) Consider the simplest price and quantity dynamics in the limit case where there are constant returns and excess demand is also a constant function. Draw the phase diagram. Suppose now that the quantity dynamics is modified by making the quantity responses depend not only on price and cost but also on the "expectation of sales," that is, on the excess demand. Will this have a stabilizing or a destabilizing effect?

17.H.3^A For $L = 3$ draw an example similar to Figure 17.H.2 but in which there is a single equilibrium that, moreover, is locally totally unstable. Could you make it a saddle?

17.I.1^A Argue that the replica procedure described at the beginning of Section 17.I does effectively include the case where the numbers of consumers of different types are not the same (assume, for simplicity, that the proportions of the different types are rational numbers). [Hint: Redefine the size of the original economy.]

17.I.2^A Consider for a one-input, one-output problem the production function $q = v^2$, where v is the amount of input. Show that the corresponding production set Y is additive but that the smallest cone containing it, Y^* , is not closed. Discuss in what sense the nonconvexity in Y is large. Argue that, whatever the number of consumers, there is no useful sense in which an equilibrium (nearly) exists.

17.I.3^B There are three commodities: the first is a high-quality good, the second is a low-quality good, and the third is labor. The first and second goods can be produced from labor according to the production functions $f_1(v) = \text{Min}\{v, 1\}$ and $f_2(v) = \text{Min}\{v^\beta, 1\}$ for $0 < \beta < 1$. The economy has one unit of labor in the aggregate. Labor has no utility value. There are two equally sized classes of agents, with a very large number of each. "Rich" and "poor" have identical endowments, but the rich own all the shares in the firms of the economy. The rich spend all their wealth on the high-quality good; the poor must buy either one quality or the other—they cannot buy both. The utility function of the poor is $u(x_1, x_2) = x_1 + \frac{1}{2}x_2$, defined for (x_1, x_2) not both positive.

(a) Which standard hypothesis of the general model does this economy fail to satisfy?

(b) Show that there can be no equilibria other than one in which both qualities of product are produced.

(c) Show that an equilibrium exists.

17.AA.1^A Consider an exchange economy in which the preferences of consumers are monotone, strictly convex, and represented by the utility functions $(u_1(\cdot), \dots, u_I(\cdot))$. Show that for any $(s_1, \dots, s_I) \gg 0$ there can be at most one Pareto optimal allocation $x = (x_1, \dots, x_I)$ such that $(u_1(x_1), \dots, u_I(x_I))$ is proportional to $(s_1, \dots, s_I)$.

17.AA.2^B Consider the welfare-theoretic approach to the equilibrium equations described in Appendix A (the Negishi approach). The existence of a solution to the system of equations $g(s) = 0$ defined there follows from a fixed-point argument similar to the one carried out in Proposition 17.C.2. Assume that you are in an exchange economy with continuous, strictly convex and strongly monotone preferences, and that $\omega_i \gg 0$ for every i . Assume also that $g(s)$ turns out to be a function rather than a correspondence (a sufficient condition for this is that preferences be representable by differentiable utility functions and that at every Pareto optimal allocation at least one consumer gets a strictly positive consumption of every good).

(a) Show that $g(s)$ is continuous.

(b) Show that $g(s)$ satisfies a sort of Walras' law: " $\sum_i g_i(s) = 0$, for every s ."

(c) Show that if $s_i = 0$ then $g_i(s) > 0$. [Hint: If $s_i = 0$ then $u_i(x_i(s)) = 0$ and so $p(s) \cdot x_i(s) = 0$.]

(d) Complete the existence proof. (Note that $g(s)$ is also defined for s with zero components. This makes matters simpler.)

17.AA.3^B Suppose that, in an exchange economy, consumption sets are $\mathbb{R}_+^L$ and preferences are representable by concave, increasing utility functions $u_i(\cdot)$. Let $\Delta = \{\lambda \in \mathbb{R}_+^L : \sum_i \lambda_i = 1\}$ be a simplex of utility weights. Suggest an equation system for Walrasian equilibrium that proceeds by associating with every λ a linear social welfare function.

17.BB.1^A Give a graphical example (for $L = 2$) of a Walrasian quasiequilibrium with strictly positive prices that is not an equilibrium for an economy in which:

- (i) For every j , $Y_j = -\mathbb{R}_+^L$.
- (ii) For every i , X_i is nonempty, closed, convex and satisfies $X_i + \mathbb{R}_+^L \subset X_i$.
- (iii) For every i , preferences are continuous, convex, and strongly monotone.
- (iv) For every i , $\omega_i \in X_i$.

Why does this example not contradict any result given in the text (see the small-type discussion after the proof of Proposition 17.BB.1)?

17.BB.2^B Consider an economy in which every consumer desires only a subset of goods and has holdings of only some goods. For the commodities desired, however, the preferences of the consumer are strongly monotone (they are also continuous) on the corresponding nonnegative orthant. Suppose in addition that $\sum_i \omega_i \gg 0$ and that the economy satisfies the following *indecomposability* condition:

It is not possible to divide consumers into two (nonempty) groups so that the consumers of one of the groups do not desire any of the commodities owned by the consumers of the other group.

Show then that any Walrasian quasiequilibrium is an equilibrium.

17.BB.3^C Consider an Edgeworth box where preferences are continuous, strictly convex and locally nonsatiated (but not necessarily monotone). Suppose also that free disposal of commodities is not possible. Argue that, nonetheless, the offer curves must cross and, therefore, that an equilibrium exists. Show that at equilibrium the two prices cannot be negative. In fact, at least one price must be positive (this is harder to show).

17.BB.4^A Prove that if (x^*, y^*, p) is a free-disposal quasiequilibrium and Y_1 satisfies free disposal, then we can get a true quasiequilibrium by changing only the production of firm 1.

17.BB.5^A Provide the missing step in the proof of Lemma 17.BB.5 (that is, show that the convexity of preferences implies that $x_i \succ_i x_{ia}$ and $x'_i \succ_i x_{ia}$ cannot both occur for $x_{ia} = \alpha x_i + (1 - \alpha)x'_i$).

17.BB.6^A Complete the proof of Lemma 17.BB.5 by verifying the convexity of $\tilde{y}_j(x, y, p)$ and of $\tilde{p}(x, y, p)$.

17.BB.7^A Complete the proof of Lemma 17.BB.6 by verifying the upper hemicontinuity of the correspondences $\tilde{y}_j(\cdot)$ and $\tilde{p}(\cdot)$.

17.BB.8^B [Existence with production externalities; see Chipman (1970) for more on this topic.] There are L goods. Good L is labor and it is the single factor of production. Consumers have consumption set $\mathbb{R}_+^L$, continuous, strongly monotone, and strictly convex preferences, and endowments only of labor. Good $\ell = 1, \dots, L - 1$ is produced in sector ℓ , which is composed of J_ℓ identical firms. The production function of a firm in sector ℓ is $f_\ell(v_\ell) = \alpha_\ell v_\ell^{\beta_\ell}$ for $0 < \beta_\ell \leq 1$. The peculiarity of the model is that the productivity coefficient α_ℓ will not be a constant but will depend on the aggregate use of labor in sector ℓ . Precisely,

$$\alpha_\ell = \gamma_\ell \left(\sum_j v_{\ell j} \right)^{\rho_\ell}, \quad \gamma_\ell > 0 \text{ and } \rho_\ell \geq 0.$$

(a) Define the notion of Walrasian equilibrium. Assume in doing so that individual firms neglect the effect on α_ℓ of their use of labor. To save on notation, suppose also that profit shares are equal across consumers.

(b) Prove the existence of a Walrasian equilibrium for the current model (make the standard additional assumptions that you find necessary). [Hint: The general proof of Appendix B needs very few adaptations.]

(c) Derive and represent the aggregate production set of each sector. Which conditions on the parameters β_ℓ , γ_ℓ , ρ_ℓ guarantee that the aggregate production set of sector ℓ exhibits increasing, constant, or decreasing returns to scale?

(d) Note that the existence conditions of (b) may be satisfied while the aggregate production set is not convex. What would happen if the externality of sector ℓ were internalized by putting all the firms of the sector under joint management?

(e) Suppose that $L = 2$, $\beta_\ell = 1$ and individual preferences are quasilinear in labor; that is, they admit a utility function $u_i(x_{1i}) + x_{2i}$. Discuss, both analytically and graphically, the bias of the equilibrium level of production relative to the social optimum.

17.BB.9^B Carry out the existence argument for the two-player-game approach described at the end of Appendix B.

Some Foundations for Competitive Equilibria

18.A Introduction

Up to this point of Part IV, the existence of markets in which prices are quoted and taken as given by economic agents has been assumed. In this chapter, we discuss four topics that, in essence, have two features in common: The first is that they all try to single out and characterize the Walrasian allocations from considerations more basic than those stated in its definition. The second is that they all emphasize the role of a large number of traders in accomplishing this task.

In Section 18.B we introduce the concept of the *core*, which can be viewed as embodying a notion of unrestricted competition. We then present the important *core equivalence theorem*.

Section 18.C examines a more restricted concept of competition: that taking place through well-specified trading mechanisms. The analysis of this section amounts to a reexamination in the general equilibrium context of the models of noncooperative competition that were presented in Section 12.F.

▪ The motivation of the remaining two sections is more normative. In Section 18.D we show how informational limitations on the part of a policy authority (constrained to use policy tools relying on *self-selection*, or *envy freeness*) may make the Walrasian allocations the only implementable Pareto optimal allocations.

In Section 18.E the objective is to characterize the Walrasian allocations, among the Pareto optimal ones, in terms of their distributional properties. In particular, we ask to what extent it can be asserted that at the Walrasian allocation everyone gets her “marginal contribution” to the collective economic well-being of society.

A number of the ideas of this chapter (especially those related to the core, but also some in Section 18.E) have come to economics from the cooperative theory of games. This therefore seems a good place to present a brief introduction to this theory; we do it in Appendix A.

18.B Core and Equilibria

The theory to be reviewed in this section was proposed by Edgeworth (1881). His aim was to explain how the presence of many interacting competitors would lead to

the emergence of a system of prices taken as given by economic agents, and consequently to a Walrasian equilibrium outcome. Edgeworth's work had no immediate impact. The modern versions of his theory follow the rediscovery of his solution concept (known now as the *core*) in the theory of cooperative games. Appendix A contains a brief introduction to the theory of cooperative games; this section, however, is self-contained. For further, and very accessible, reading on the material of this section, we refer to Hildenbrand and Kirman (1988).

The theory of the core is distinguished by its parsimony. Its conceptual apparatus does not appeal to any specific trading mechanism nor does it assume any particular institutional setup. Informally, the notion of competition that the theory explores is one in which traders are well informed of the characteristics (endowments and preferences) of other traders, and in which the members of any group of traders can bind themselves to any mutually advantageous agreement. The simplest example is a buyer and a seller exchanging a good for money, but we can also have more complex arrangements involving many individuals and goods.

Formally, we consider an economy with I consumers. Every consumer i has consumption set $\mathbb{R}_+^L$, and endowment vector $\omega_i \geq 0$, and a continuous, strictly convex, strongly monotone preference relation $\succsim_i$. There is also a publicly available constant returns convex technology $Y \subset \mathbb{R}^L$.¹ For example, we could have $Y = -\mathbb{R}_+^L$, that is, a pure exchange economy. All of these assumptions are maintained for the rest of the section.

As usual, we say that an allocation $x = (x_1, \dots, x_I) \in \mathbb{R}_+^{LI}$ is *feasible* if $\sum_i x_i = y + \sum_i \omega_i$ for some $y \in Y$.

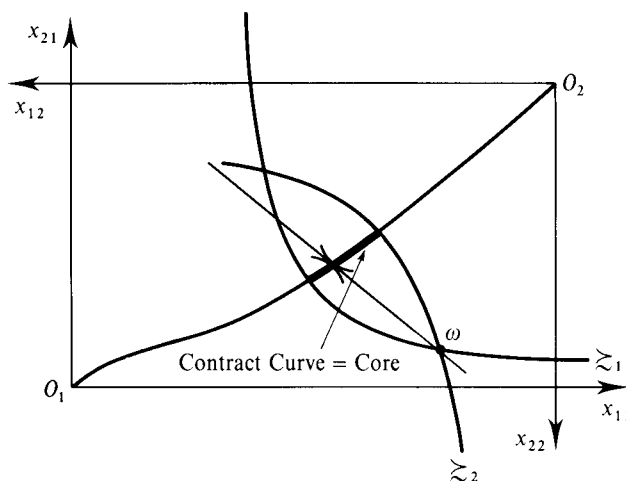
With a slight abuse of notation, we let the symbol I stand for both the number of consumers and the *set* of consumers. Any nonempty subset of consumers $S \subset I$ is then called a *coalition*. Central to the concept of the core is the identification of circumstances under which a coalition of consumers can reach an agreement that makes every member of the coalition better off. Definition 18.B.1 provides a formal statement of these circumstances.

Definition 18.B.1: A coalition $S \subset I$ *improves upon*, or *blocks*, the feasible allocation $x^* = (x_1^*, \dots, x_I^*) \in \mathbb{R}_+^{LI}$ if for every $i \in S$ we can find a consumption $x_i \geq 0$ with the properties:

- (i) $x_i \succ_i x_i^*$ for every $i \in S$.
- (ii) $\sum_{i \in S} x_i \in Y + \{\sum_{i \in S} \omega_i\}$.

Definition 18.B.1 says that a coalition S can improve upon a feasible allocation x^* if there is some way that, by using *only* their endowments $\sum_{i \in S} \omega_i$ and the publicly available technology Y , the coalition can produce an aggregate commodity bundle that can then be distributed to the members of S so as to make each of them better off.

1. The constant returns assumption is important. With general production sets the difficulty is that we cannot avoid being explicit about ownership shares. However, these have been defined to be *profit* shares, which makes our conceptual apparatus dependent on the very notion of prices whose emergence we are currently trying to explain. Thus we stick here to the case of constant returns. This is not a serious restriction: recall from Section 5.B (Proposition 5.B.2) that it is always possible to reduce general technologies to the constant returns case by reinterpreting the ownership shares as endowments of an additional "managerial" input.

**Figure 18.B.1**

The core equals the contract curve in the two-consumer case.

Definition 18.B.2: We say that the feasible allocation $x^* = (x_1^*, \dots, x_I^*) \in \mathbb{R}_+^{LI}$ has the *core property* if there is no coalition of consumers $S \subset I$ that can improve upon x^* . The *core* is the set of allocations that have the core property.

We can see in the Edgeworth box of Figure 18.B.1 that for the case of two consumers the core coincides with the *contract curve*. With two consumers there are only three possible coalitions: $\{1, 2\}$, $\{1\}$, and $\{2\}$. Any allocation that is not a Pareto optimum will be blocked by coalition $\{1, 2\}$.² Any allocation in the Pareto set that is not in the contract curve will be blocked by either $\{1\}$ or $\{2\}$. With more than two consumers there are other potential blocking coalitions, but the fact that the coalition of the whole is always one of them means that *all allocations in the core are Pareto optimal*.

We also observe in Figure 18.B.1 that the Walrasian equilibrium allocations, which belong to the contract curve, have the core property. Proposition 18.B.1 tells us that this is true with complete generality. The proposition amounts to an extension of the first welfare theorem. Indeed, in the current terminology, the first welfare theorem simply says that a Walrasian equilibrium cannot be blocked by the coalition of the whole.³ The following result, Proposition 18.B.1, shows that it also cannot be blocked by any other coalition.

Proposition 18.B.1: Any Walrasian equilibrium allocation has the core property.

Proof: We simply duplicate the proof of the first welfare theorem (Proposition 16.C.1). We present it for the exchange case. See Exercise 18.B.1 for the case of a general constant returns technology.

Let $x^* = (x_1^*, \dots, x_I^*)$ be a Walrasian allocation with corresponding equilibrium

2. With continuity and strong monotonicity of preferences, if a feasible allocation is Pareto dominated, then it is Pareto dominated by a feasible allocation that strictly improves the utility of *every* consumer. To accomplish this we simply transfer a very small amount of any good from the consumer that is made better off to every other consumer. If the amount transferred is sufficiently small then, by the continuity of preferences, the transferring consumer is still better off, while, by strong monotonicity, every other consumer is made strictly better off.

3. Keep in mind the point made in footnote 2.

The ability to do this depends, of course, on the way we have drawn the indifference curves. Nonetheless, as we will see, we are always able to form a blocking coalition of this sort if we have sufficiently many consumers of each type.

The version of the core equivalence theorem that we will present is in essence the original of Edgeworth, as generalized by Debreu and Scarf (1963). It builds on the intuition we have just discussed.

To begin, let the set $H = \{1, \dots, H\}$ stand for a set of *types* of consumers, with each type h having preferences $\succeq_h$ and endowments ω_h . For every integer $n > 0$, we then define the N -replica economy as an economy composed of N consumers of each type, for a total number of consumers $I_N = NH$.

We refer to the allocations in which consumers of the same type get the same consumption bundles as *equal-treatment allocations*. Proposition 18.B.2 shows that any allocation in the core must be an equal-treatment allocation. (We hasten to add that this is true for the current replica structure, where there are equal numbers of consumers of each type. It does not hold in general; see Exercise 18.B.2.)

Proposition 18.B.2: Denoting by hn the n th individual of type h , suppose that the allocation

$$x^* = (x_{11}^*, \dots, x_{1n}^*, \dots, x_{1N}^*, \dots, x_{H1}^*, \dots, x_{Hn}^*, \dots, x_{HN}^*) \in \mathbb{R}_+^{LHN}$$

belongs to the core of the N -replica economy. Then x^* has the *equal-treatment property*, that is, all consumers of the same type get the same consumption bundle:

$$x_{hm}^* = x_{hn}^* \quad \text{for all } 1 \leq m, n \leq N \text{ and } 1 \leq h \leq H.$$

Proof: Suppose that the feasible allocation $x = (x_{11}, \dots, x_{HN}) \in \mathbb{R}_+^{LHN}$ does not have the equal-treatment property because, say, $x_{1m} \neq x_{1n}$ for some $m \neq n$. We show that x does not have the core property. In particular, we claim that x can be improved upon by any coalition of H members formed by choosing from every type a worst-treated individual among the consumers of that type. Suppose without loss of generality that, for every h , consumer $h1$ is one such worse-off individual, that is, $x_{hn} \succeq_h x_{h1}$ for all h and n . Define now the average consumption for each type: $\hat{x}_h = (1/N) \sum_n x_{hn}$. By the strict convexity of preferences we have (recall that consumers of type 1 are not treated identically)

$$\hat{x}_h \succeq_h x_{h1} \quad \text{for all } h \quad \text{and} \quad \hat{x}_1 \succ_1 x_{11}. \quad (18.B.1)$$

We claim that the coalition $S = \{11, \dots, h1, \dots, H1\}$, formed by H members, can attain by itself the consumptions $(\hat{x}_1, \dots, \hat{x}_H) \in \mathbb{R}_+^{LH}$. Therefore, by (18.B.1), the original nonequal-treatment allocation can be blocked by S .⁵ To check the feasibility of $(\hat{x}_1, \dots, \hat{x}_H) \in \mathbb{R}_+^{LH}$ for S , note that, because of the feasibility of $x = (x_{11}, \dots, x_{HN}) \in \mathbb{R}_+^{LHN}$, there is $y \in Y$ such that $\sum_h \sum_n x_{hn} = y + N(\sum_h \omega_h)$, and therefore

$$\sum_h \hat{x}_h = \frac{1}{N} \sum_h \left(\sum_n x_{hn} \right) = \frac{1}{N} y + \sum_h \omega_h.$$

5. Recall that preferences are strongly monotone and continuous, so that if S can achieve an allocation that does strictly better than x^* for some of its members, and at least as well as x^* for all of them, then it can also achieve an allocation that does strictly better for all of its members.

But by the constant returns assumption on Y , $(1/N)y \in Y$ and so we conclude that $(\hat{x}_1, \dots, \hat{x}_H) \in \mathbb{R}_+^{LH}$ is feasible for coalition S . ■

Proposition 18.B.2 allows us to regard the core allocations as vectors of fixed size LH , irrespective of the replica that we are concerned with. As a matter of terminology, we call a vector $(x_1, \dots, x_H) \in \mathbb{R}_+^{LH}$ a *type allocation* and, for any replica N , interpret it as the equal-treatment allocation to consumers where each consumer of type h gets x_h . A type allocation $(x_1, \dots, x_H) \in \mathbb{R}_+^{LH}$ is feasible if $\sum_h x_h = y + \sum_h \omega_h$ for some $y \in Y$. Note that for any replica N the corresponding equal-treatment allocation is feasible because

$$\sum_h Nx_h = Ny + N\left(\sum_h \omega_h\right)$$

and $Ny \in Y$ by the constant returns assumption on Y .

By Proposition 18.B.2 the core allocations of a replica economy can be viewed as feasible type allocations. Define by $C_N \subset \mathbb{R}_+^{LH}$ the set of feasible type allocations for which the equal-treatment allocations induced in the N -replica have the core property. Note that C_N does depend on N . Nonetheless, we always have $C_{N+1} \subset C_N$ because a type allocation blocked in the N -replica will be blocked also in the $(N+1)$ -replica by a coalition having exactly the same composition as the one that blocked in the N -replica. Thus, as a subset of $\mathbb{R}_+^{LH}$ the core can only get smaller when $N \rightarrow \infty$. At the same time, we know from Proposition 18.B.1 that the core cannot vanish because the Walrasian equilibrium allocations belong to C_N for all N . More precisely, the set of Walrasian type allocations is independent of N (see Exercise 18.B.3) and contained in all C_N . The core equivalence theorem (which, in the current replica context, is the formal term for the combination of Propositions 18.B.1, 18.B.2 and the forthcoming Proposition 18.B.3) asserts that the Walrasian equilibrium allocations are the only surviving allocations in the core when $N \rightarrow \infty$.

Proposition 18.B.3: If the feasible type allocation $x^* = (x_1^*, \dots, x_H^*) \in \mathbb{R}_+^{LH}$ has the core property for all $N = 1, 2, \dots$, that is, $x^* \in C_N$ for all N , then x^* is a Walrasian equilibrium allocation.

Proof: To make the proof as intuitive as possible we restrict ourselves to a special case: a pure exchange economy in which, for every h , $\succsim_h$ admits a continuously differentiable utility representation $u_h(\cdot)$ [with $\nabla u_h(x_h) \gg 0$ for all x_h]. In addition, the initial endowments vector ω_h is preferred to any consumption x_h that is not strictly positive. This guarantees that any core allocation is interior. We emphasize that these simplifying assumptions are not required for the validity of the result.

Suppose that $x = (x_1, \dots, x_H) \in \mathbb{R}_+^{LH}$ is a feasible type allocation that is not a Walrasian equilibrium allocation. Our aim is to show that if N is large enough then x can be blocked.

We may as well assume that x is Pareto optimal (otherwise the coalition of the whole blocks and we are done) and that $x_h \gg 0$ (otherwise a consumer of type h alone could block). Because of Pareto optimality we can apply the second welfare theorem (Proposition 16.D.1) and conclude that x is a price equilibrium with transfers with respect to some $p = (p_1, \dots, p_L)$. If x is not Walrasian then there must be some h , say $h = 1$, with $p \cdot (x_1 - \omega_1) > 0$. Informally, type 1 receives a positive net transfer from the rest of the economy and is thus relatively favored (interpretatively, think

of type 1 as the most favored). We shall show that, as long as N is large enough, it would pay for the members of all the other types in the economy to form a coalition with $N - 1$ consumers of type 1 (i.e., to throw out one consumer of type 1).

More precisely, if a member of type 1 is eliminated then to attain feasibility the rest of the economy must absorb her net trade $x_1 - \omega_1$. That, of course, presents no difficulty for the positive entries (those commodities for which the rest of the economy is a net contributor to this consumer of type 1), but it is not so simple for the negative ones (the commodities where the rest of the economy is the net beneficiary). The most straightforward methodology is to simply distribute the gains and losses equally. In summary, our coalition is formed by $(N - 1) + N(H - 1)$ members and, for every type h , every member of type h gets

$$x'_h = x_h + \frac{1}{(N - 1) + N(H - 1)} (x_1 - \omega_1).$$

Note that

$$\begin{aligned} (N - 1)x'_1 + Nx'_2 + \cdots + Nx'_H &= (N - 1)x_1 + Nx_2 + \cdots + Nx_H + (x_1 - \omega_1) \\ &= N\omega_1 + \cdots + N\omega_H - x_1 + x_1 - \omega_1 \\ &= (N - 1)\omega_1 + N\omega_2 + \cdots + N\omega_H. \end{aligned}$$

Hence, the proposed consumptions are feasible for the proposed coalition. Note also that the consumptions are nonnegative if N is large enough. For every h , every consumer of type h in the coalition moves from x_h to x'_h . Is this an improvement or a loss? The answer is that if N is large enough then it is an unambiguous gain. To see this, observe that $p \cdot (x_1 - \omega_1) > 0$ implies $\nabla u_h(x_h) \cdot (x_1 - \omega_1) > 0$ for every h because p and $\nabla u_h(x_h)$ are proportional. As we can then see in Figure 18.B.3 (or, analytically, from Taylor's formula; see Exercise 18.B.4) there is $\bar{\alpha} > 0$ with the property that, for every h , $u_h(x_h + \alpha(x_1 - \omega_1)) > u_h(x_h)$ whenever $0 < \alpha < \bar{\alpha}$. Hence, for any N with $(1/[(N - 1) + N(H - 1)]) < \bar{\alpha}$ the coalition will actually be blocking.

Intuitively, we have done the following. The coalition needs to absorb $x_1 - \omega_1$. Evaluated at the marginal shadow prices of the economy, this is a favorable "project" for the coalition since $p \cdot (x_1 - \omega_1) > 0$. If the coalition is numerous then we can make sure that every member will have to absorb only a very small piece of the project. Hence the individual portions of the project will all be "at the margin"

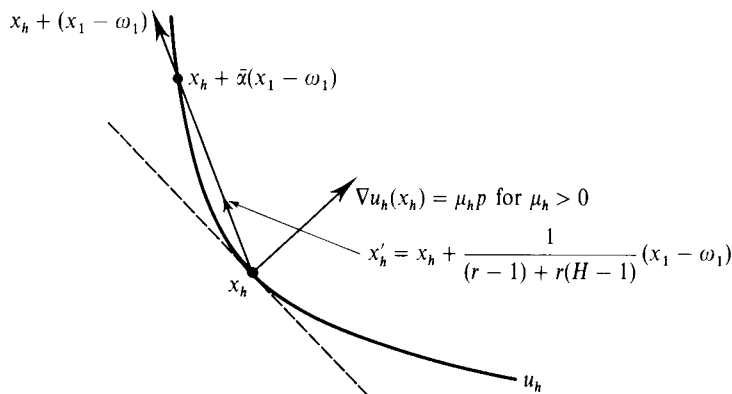


Figure 18.B.3

The consumption change of a consumer of type h in the blocking coalition.

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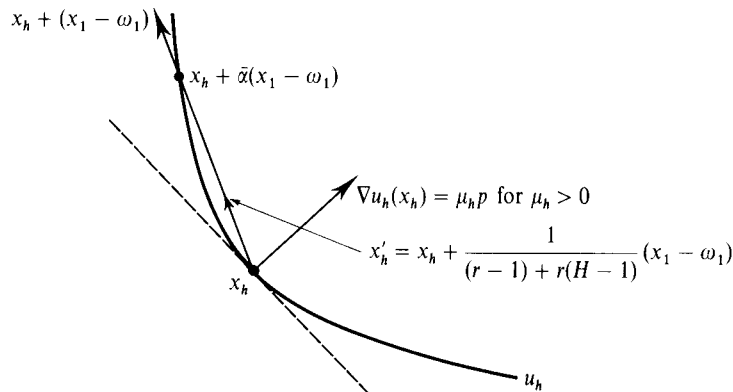


Figure 18.B.3

The consumption change of a consumer of type h in the blocking coalition.

and, therefore, will also be individually favorable (recall Section 3.I for similar arguments).⁶ ■

We saw in Proposition 18.B.1 that the half of the core equivalence theorem that asserts that Walrasian allocations have the core property generalizes the first welfare theorem. In its essence, the half asserting that, provided the economy is large, core allocations are Walrasian constitutes a version of the second welfare theorem. To understand this it may be useful to go back to the general (nonreplica) setup and formulate the property of a core allocation being Walrasian in terms of the existence of a price support for a certain set. For simplicity, we restrict ourselves to the pure exchange case.

Given a core allocation $x = (x_1^*, \dots, x_I^*) \in \mathbb{R}_+^{LI}$ then, in analogy with the construction used in the proof of the second welfare theorem (Proposition 16.D.1) we can define the sets

$$V_i = \{x_i : x_i \succ_i x_i^*\} \cup \{\omega_i\} \subset \mathbb{R}^L$$

$$V = \sum_{i \in I} V_i \subset \mathbb{R}^L$$

We have $\sum_i \omega_i \in V$. But there is more: *the core property for x^* implies that $\sum_i \omega_i$ belongs to the boundary of V* . To see this, note that if $\sum_i \omega_i$ is in the interior of V then there is $z \in V$ such that $z \ll \sum_i \omega_i$; that is, there is $x' = (x'_1, \dots, x'_I)$ with $x'_i \in V_i$ for every i and $\sum_i x'_i = z \ll \sum_i \omega_i$. Hence, x' is feasible, $x' \neq (\omega_1, \dots, \omega_I)$, and, for every i , either $x'_i \succ_i x_i^*$ or $x'_i = \omega_i$. It follows that the set of consumers $S = \{i : x'_i \neq \omega_i\}$ is nonempty, that $x'_i \succ_i x_i^*$ for every $i \in S$, and that

$$\sum_{i \in S} x'_i \ll \sum_{i \in I} \omega_i - \sum_{i \notin S} x'_i = \sum_{i \in I} \omega_i - \sum_{i \notin S} \omega_i = \sum_{i \in S} \omega_i.$$

Thus S is a blocking coalition.

The next claim is that if $p = (p_1, \dots, p_L) \neq 0$ supports V at $\sum_i \omega_i$, that is, $p \cdot z \geq p \cdot (\sum_i \omega_i)$ for all $z \in V$, then p must be a Walrasian price vector for $x^* = (x_1^*, \dots, x_I^*)$. To verify this, note first that, for every i , we have $x'_i \succ_i x_i^*$ for some x'_i arbitrarily close to x_i^* . Therefore, $x'_i + \sum_{k \neq i} \omega_k \in V$ and so $p \cdot (x'_i + \sum_{k \neq i} \omega_k) \geq p \cdot (\omega_i + \sum_{k \neq i} \omega_k)$. Going to the limit (i.e., letting $x'_i \rightarrow x_i^*$), this yields $p \cdot x_i^* \geq p \cdot \omega_i$ for all i . Because $\sum_i x_i^* \leq \sum_i \omega_i$, we must therefore have $p \cdot x_i^* = p \cdot \omega_i$ for all i . In addition, whenever $x'_i \succ_i x_i^*$ we have $p \cdot (x'_i + \sum_{k \neq i} \omega_k) \geq p \cdot (\omega_i + \sum_{k \neq i} \omega_k)$ and so $p \cdot x'_i \geq p \cdot \omega_i$. If we exploit the continuity and strong monotonicity of preferences as we did in Section 16.D (or in Appendix B of Chapter 17), we can strengthen the last conclusion to $p \cdot x'_i > p \cdot \omega_i$.

The key difference from the case of the second welfare theorem (studied in Section 16.D) is that $V \subset \mathbb{R}^L$ does not need to be convex and that therefore a nonzero $p \in \mathbb{R}_+^L$ supporting V at $\sum_i \omega_i$ may not exist. The reason for the lack of convexity is that the individual sets $V_i \subset \mathbb{R}^L$ need not be convex: V_i is the union of the preferred set at x_i^* , which is convex, and the initial endowment vector ω_i , which will typically be outside this preferred set and therefore disconnected from it. However, *if the (possibly nonconvex) sets $V_i \subset \mathbb{R}^L$ being added are numerous, then the sum $\sum_i V_i \subset \mathbb{R}^L$ is "almost" convex*. Thus, the existence of (almost) supporting prices for core allocations can be seen as yet another instance of the convexifying effects of aggregation.

We end by mentioning an elegant approach to core theory pioneered by Aumann (1964) and Vind (1964). It consists of looking at a model where there is an actual continuum of consumers and where we replace all the summations by integrals. The beauty of the approach is that all the approximate results then hold exactly. The core equivalence theorem, for example,

6. See Anderson (1978) for a different line of proof that makes minimal assumptions on the economy.

takes the form: An allocation belongs to the core if and only if it is a Walrasian equilibrium allocation.

18.C Noncooperative Foundations of Walrasian Equilibria

The idea of competition that underlies the theory of the core is very unstructured; there are no trading institutions and, in principle, any conceivable profitable opportunity can be taken advantage of. It is because of this, for example, that core allocations are guaranteed to be Pareto optimal.

In many economic applications, however, the structure of competition is given. Trade takes place through some type of market mechanism making explicit use of prices. The set of instruments and the information available to competitors are then limited. Yet, we also expect that price taking will emerge if individual competitors are small relative to the size of the market. We have already investigated this topic in Section 12.F. We reexamine it here because there are a few general equilibrium qualifications worth taking into account.

There are many models of price-mediated competition arising in applications. We will describe three of them, but before doing so, we present an abstract treatment emphasizing the main issues.

Suppose there are I economic agents (abstract competitors, perhaps firms). There are also a set $P \subset \mathbb{R}^L$ of possible price vectors and a set A of "market actions." Every i has a set $A_i \subset A$ and an endowment vector $\omega_i \in \mathbb{R}^L$. For every $a_i \in A_i$ and $p \in P$, a *trading rule* defined on $A \times P$ and with values in $\mathbb{R}^L$, assigns a net trade vector $g(a_i; p)$ to agent i , satisfying $p \cdot g(a_i; p) = 0$. Given an array $a = (a_1, \dots, a_I)$ of actions, there is then a market clearing process that generates a price vector $p(a) \in P$. We also assume that every i has a utility function $u_i(g(a_i; p) + \omega_i)$, thus indirectly defined on $A_i \times P$.

The previous setup suggests treating the problem by the methodology of noncooperative games, as presented in Chapter 8.

Definition 18.C.1: The profile of actions $a^* = (a_1^*, \dots, a_I^*) \in A_1 \times \dots \times A_I$ is a *trading equilibrium* if, for every i ,

$$u_i(g(a_i^*; p(a^*)) + \omega_i) \geq u_i(g(a_i; p(a_i, a_{-i}^*)) + \omega_i) \quad \text{for all } a_i \in A_i.^7$$

The concept of noncooperative equilibrium incorporated in Definition 18.C.1 is the same as the one we used in Chapters 8 and 12. As there, and in contrast to the analysis of the core in Section 18.B, such equilibria need not be Pareto optimal. The question we now pose ourselves is: Under what conditions is it the case that, if individual traders are small relative to the size of the economy, the system of markets approximates a price-taking environment in which, effectively, every trader optimizes given a competitive budget set (and in which, therefore, the equilibria will be nearly Pareto optimal).

7. As it has become customary, we follow the notation $(a_i, a_{-i}^*) = (a_1^*, \dots, a_{i-1}^*, a_i, a_{i+1}^*, \dots, a_I^*)$.

Given $a = (a_1, \dots, a_I) \in A_1 \times \dots \times A_I$, define

$$B_i(a) = \{x_i \in \mathbb{R}_+^L : x_i - \omega_i \leq g(a'_i; p(a'_i, a_{-i})) \text{ for some } a'_i \in A_i\}$$

as the *effective budget set* of trader i at a . In words, $B_i(a)$ is the set of net trades that trader i can achieve through *some* choice of a_i , given that the remaining traders are choosing a_{-i} . This set of achievable net trades will be close to the Walrasian budget

$$B(p(a), p(a) \cdot \omega_i) = \{x_i \in \mathbb{R}_+^L : p(a) \cdot x_i \leq p(a) \cdot \omega_i\}$$

if the following two types of conditions both hold:

(1) *Insensitivity of prices to own actions.* For the boundary of $B_i(a)$ to be (almost) contained in a hyperplane, we need the price-clearing function $p(a_i, a_{-i})$ to be very insensitive to a_i .⁸ Often this will be guaranteed if the economy is large and, consequently, every competitor is of small relative size. Suppose, for example, that $p(a)$ has the form $p((1/r) \sum_i a_i)$ where r is a size parameter (perhaps the number of consumers). It is actually quite common that the problem be given as, or can be transformed into, one in which actions affect price only through some average. At any rate, if this is the case, then the essential fact is clear: *As long as $p(\cdot)$ depends continuously on the average action $(1/r) \sum_i a_i$ the dependence of prices on individual actions (assume the A_i are bounded) will become negligible as the economy becomes large; that is, as $r \rightarrow \infty$.* Continuity of $p(\cdot)$ is therefore a key property.

(2) *Individual spanning.* Even if $p(a)$ is practically independent of the actions of an individual, we could still have a failure of *individual spanning*. That is, the boundary of the set $B_i(a)$, while flat, may be “too short,” as in Figure 18.C.1(a), or even lower-dimensional, as in Figure 18.C.1(b) where it reduces to the initial endowment vector (no trade at all is possible). Individual spanning will have to be checked in every case. Verifying it will typically involve showing, first, that $g(a_i, p)$ is sensitive enough to a_i and, second, that A_i is large enough.

We now briefly, and informally, discuss three examples illustrating these ideas.⁹

Example 18.C.1: General Equilibrium, Single-Good Cournot Competition. This is in essence the same model studied in Section 12.C, except that we now admit a completely general form of “inverse demand function,” that is, of the correspondence that assigns market-clearing prices to aggregate production decisions. This is meant to reflect the possibility of wealth effects (a hallmark of the general equilibrium approach).

To be concrete, suppose that we have two goods: the first is a consumption good and the second is “money” (which is also the unit of account with price equal to 1). There are r identical consumers, each endowed with a unit of money. For a price $p \in \mathbb{R}$ of the consumption good the demand of a consumer for this good is $x(p) \in \mathbb{R}$. There are also J firms producing the consumption good out of money. Firms set quantities. Marginal cost, up to a unit of capacity, is zero. To minimize complexity,

8. Recall that $p(a'_i, a_{-i}) \cdot g(a'_i; p(a'_i, a_{-i})) = 0$ for all a'_i . Therefore, if $p(a'_i, a_{-i})$ is (almost) independent of a'_i then $B_i(a)$ is (almost) contained in the hyperplane perpendicular to $p(a)$.

9. For more on general equilibrium Cournotian models in the style of Examples 18.C.1 and 18.C.2, see Gabszewicz and Vial (1972) and Novshek and Sonnenschein (1978). For a survey of the general area, see Mas-Colell (1982).

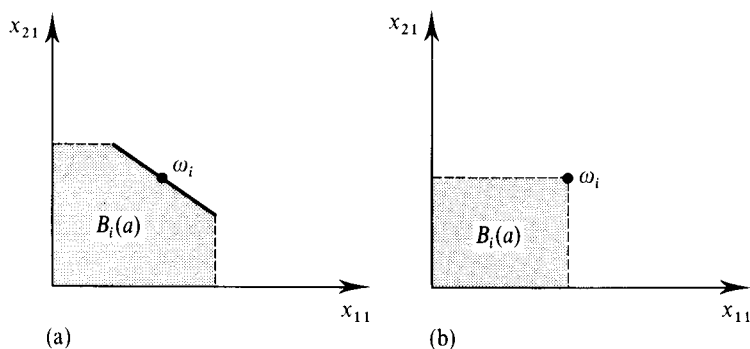


Figure 18.C.1

(a) and (b): Two nonspanning effective budget sets.

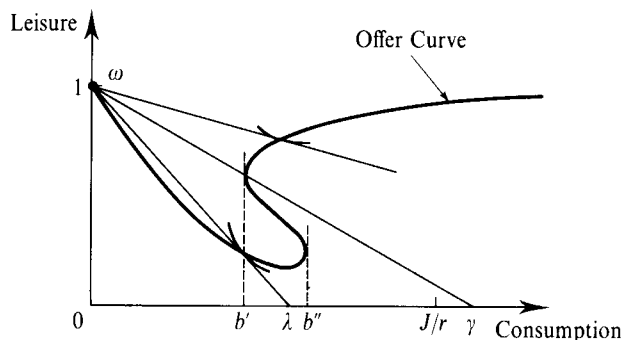


Figure 18.C.2

An economy with no continuous price selection.

assume that the owners of the firms are a separate group of agents interested only in the consumption of money. Therefore, for any total production $q = \sum_j q_j$ and size parameter r , the market price of the consumption good must solve the general equilibrium system $rx(p) = q$, or $x(p) = q/r$. We suppose that for any q/r the market selects a solution $p(q/r)$ to this equation.¹⁰

In the quasilinear, partial equilibrium context of Chapter 12, $x(\cdot)$ is a decreasing continuous function and therefore its inverse $p(\cdot)$ exists and is *continuous* (and decreasing). It then follows that, when r is large, $p(\sum_j q_j/r)$ is quite insensitive to the decision of any particular firm. Hence, firms are almost price takers and, as a consequence, the Cournot equilibria are almost Walrasian. Yet in the current general equilibrium context, $p(\cdot)$ may be unavoidably discontinuous. This is illustrated in Figure 18.C.2, where we represent the offer curve of a consumer. In the figure, there is no way to select money demands, and therefore prices, continuously over the offer curve as consumption per capita q/r ranges from 0 to J/r .¹¹ The location of potential Cournot equilibria will depend on how the market selects $p(\cdot)$ in the domain $[b', b'']$ of consumptions per capita, but the possibility of Cournot equilibria bounded away from the Walrasian equilibrium irrespective of the size of the economy is quite real. A particular price selection $p(\cdot)$ has been chosen in Figure 18.C.3. Note, first, that at the Walrasian equilibria of this model we must have every firm producing at capacity (and so the Walrasian equilibrium price is p^*). Yet, provided $r > (J\lambda/\gamma b')$,

10. To view this example as a particular case of the abstract trading model described above, you should think of the J firms as the players. Firm j is "endowed" with a unit of the first good and its strategy variable is $q_j \in [0, 1] = A_j$. Finally, the trading rule is $g(q_j; p, 1) = (-q_j, pq_j)$.

11. The example is contrived in that "money" is a Giffen good. If consumers were not identical, this feature would not be required.

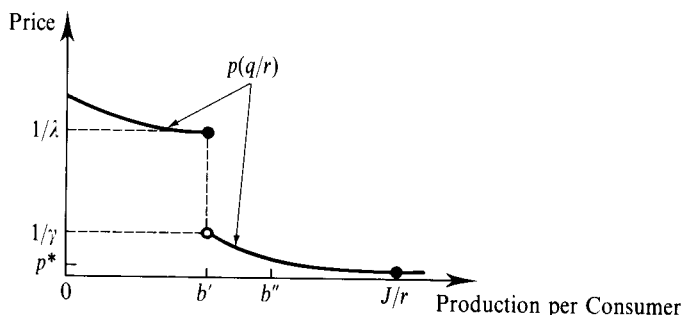


Figure 18.C.3
A price equilibrium selection.

every firm producing $rb'/J < 1$ (for a consumption per capita of b') constitutes a Cournot equilibrium: because $p(\cdot)$ is very elastic in the domain $[0, b']$ it will not pay any firm to contract production; and if any firm expands production, no matter how slightly, a precipitous and unprofitable drop in prices ensues.¹² [See Roberts (1980) for more on this point.] ■

Example 18.C.2: Cournot Competition among Complements. We modify the previous example in only two respects: (1) There are two consumption goods; (2) firms are producers of either the first or the second of these. The respective number of firms is J_1 and J_2 . To be very simple we assume that the consumer has a quasilinear utility with money as numeraire. If the concave, strictly increasing utility function for the two consumption goods is $\psi(x_1, x_2)$ then, for any total productions (q_1, q_2) , market clearing prices are

$$p(q_1, q_2) = \nabla \psi(q_1/r, q_2/r) = \left(\frac{\partial \psi(q_1/r, q_2/r)}{\partial x_1}, \frac{\partial \psi(q_1/r, q_2/r)}{\partial x_2} \right) \gg 0.$$

The Walrasian equilibrium productions are (J_1, J_2) . Suppose, now, to take an extreme situation, that the two consumption goods are complements in the sense that the consumption of one is absolutely necessary for the enjoyment of the other, or $\psi(0, x_2) = \psi(x_1, 0) = 0$ for any x_1 and x_2 . Then we claim that lack of activity (i.e., the null production of the two goods) is an equilibrium. The reason is clear enough: If $q_2 = 0$ then any positive supply $q_1 > 0$ of good one can only be absorbed by the market at $p_1 = \nabla_1 \psi(q_1/r, 0) = 0$. Thus, no firm has an incentive to produce any amount of good 1 (and similarly for good 2). Economically, the difficulty is that the cooperation of at least two firms is needed to activate a market. Technically, we have a failure of continuity of clearing prices at $(0, 0)$ since $p(\varepsilon, 0) = 0$ for all $\varepsilon > 0$ but the limit of $p(\varepsilon, \varepsilon)$ as ε goes to zero remains bounded away from zero.¹³ ■

Example 18.C.3: Trading Posts. This example belongs to a family proposed by Shapley and Shubik (1977). It is not particularly realistic but it has at least three

12. When every firm produces rb'/J , the profits of one firm are $rb'/J\lambda > 1/\gamma$. But $1/\gamma$ is an upper bound for the profits of any firm that deviates from the suggested production by producing more. Hence an output level of rb'/J for every firm constitutes an equilibrium.

13. The complementarity makes it impossible for ψ to be continuously differentiable at the origin. Therefore, $p(\cdot)$ fails to be continuous. This is the crucial aspect for the example. Note that discontinuity at the origin is a natural occurrence: it will arise, for example, whenever the indifference map of $\psi(\cdot)$ is homothetic (but not linear). See Hart (1980) for more on this issue.

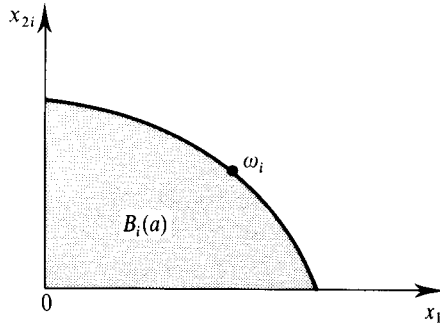


Figure 18.C.4
An effective budget set
for the trading post
Example 18.C.3.

virtues: it constitutes a complete general equilibrium model, all of the participants interact strategically (in the two previous examples, consumers adjust passively), and it is analytically simple to manipulate.

There are L goods and I consumers. Consumer i has endowment $\omega_i \in \mathbb{R}_+^L$. The L th commodity, to be called “money,” is treated asymmetrically. For each of the first $L - 1$ goods there is a *trading post* exchanging money for the good. At each trading post $\ell \leq L - 1$, each consumer i can place nonnegative bids $a_{\ell i} = (a'_{\ell i}, a''_{\ell i}) \in \mathbb{R}_+^2$. The interpretation is that an amount $a'_{\ell i}$ of good ℓ is placed at the offer side of the trading post to be exchanged for money. Similarly, an amount $a''_{\ell i}$ of money is placed in the demand side to be exchanged for good ℓ . Accordingly, the bids are also constrained by $a'_{\ell i} \leq \omega_{\ell i}$ and $\sum_{\ell \leq L-1} a''_{\ell i} \leq \omega_{Li}$.

Given the bids of consumer i in the trading posts $\ell \leq L - 1$ and prices $(p_1, \dots, p_{L-1}, 1)$ the mechanism is completed by the trading rule:

$$g_{\ell}(a_{1i}, \dots, a_{L-1,i}; p_1, \dots, p_{L-1}, 1) = \frac{a''_{\ell i}}{p_{\ell}} - a'_{\ell i}$$

for all $\ell < L - 1$. The trade for the money good is derived from the budget constraint of the consumer.

Given a vector $a = (a_{11}, \dots, a_{L-1,1}, \dots, a_{1I}, \dots, a_{L-1,I})$ of bids for all consumers, the clearing prices in terms of money are determined as the ratio of the amount of money offered to the amount of good offered:

$$p_{\ell}(a) = \frac{\sum_i a''_{\ell i}}{\sum_i a'_{\ell i}} \quad \ell = 1, \dots, L - 1. \quad (18.C.1)$$

Note that $p_{\ell}(a)$ is well defined and continuous *except* when there are no offers at the trading post ℓ [i.e., except when $a'_{\ell i} = 0$ for all i].¹⁴

A typical effective budget set for agent i is convex and, provided that $\sum_{k \neq i} a'_{\ell k} \neq 0$ and $\sum_{k \neq i} a''_{\ell k} \neq 0$ for all $\ell \leq L - 1$, it has an upper boundary containing no straight segments (you are asked to formally verify this in Exercise 18.C.1). This reflects the fact that as a consumer increases her bid in one side of a market the terms of trade turn against her. Figure 18.C.4 gives an illustration for the case $L = 2$.

14. For the special, but important, case in which there is a single trading post (i.e., $L = 2$), we can go a bit farther. When $\sum_i a''_{\ell i} > 0$ and $\sum_i a'_{\ell i} = 0$, the relative price of money is still well defined: it is zero. The essential difficulty in defining relative prices arises when $\sum_i a'_{\ell i} = 0$ and $\sum_i a''_{\ell i} = 0$.

It follows from expression (18.C.1) that approximate price taking will prevail in any trading post that is *thick* in the sense that the aggregate positions taken on the two sides of the market are large relative to the size of the initial endowments of any consumer. A necessary condition for thickness is that there be many consumers. But this is not sufficient: it is possible even in a large economy to have equilibrium where some market is *thin* and, as a consequence, a trading equilibrium may be far from a Walrasian equilibrium. In fact, any trading equilibrium for a model where a trading post ℓ is closed (i.e., the trading post does not exist) will remain an equilibrium if the trading post is open but stays inactive. That is, if we put $a_{\ell i} = (a'_{\ell i}, a''_{\ell i}) = 0$ for all i . Economically, this is related to Example 18.C.2: it takes at least two agents (here a buyer and a seller) to activate a market. Mathematically, the difficulty is again the impossibility of assigning prices continuously when $a_{\ell i} = 0$ for all i .

Up to now, in this and previous examples, all of the instances of trading equilibria not approaching a Walrasian outcome when individual competitors are small have been related to failures of continuity of market equilibrium prices. But the current example also lends itself to illustration of the individual spanning problem. Indeed, even if markets are thick and therefore prices, from the individual point of view, are almost fixed, it remains true that the trading post structure imposes the restriction that *goods can only be exchanged for money on hand* (in macroeconomics this restriction is called the cash-in-advance, or the Clower, constraint). Money obtained by selling goods cannot be applied to buy goods. Therefore, for a given individual the Walrasian budget set will be (almost) attainable only if the initial endowments of money are sufficient, that is, only if at the solution of the individual optimization problem the constraint $\sum_{\ell \leq L-1} a''_{\ell i} \leq \omega_{Li}$ is not binding. But there is no general reason why this should be so. Suppose, to take an extreme case, that $\omega_{Li} = 0$. Then consumer i simply cannot buy goods at all. ■

18.D The Limits to Redistribution

In Section 16.D we saw that, under appropriate convexity conditions and provided that wealth can be transferred in a lump-sum manner, Pareto optimal allocations can be supported by means of prices. However, as we also pointed out there, a necessary condition for lump-sum payments to be possible is the ability of the policy authority to tell who is who—that is, to be able to precisely identify the characteristics (preferences and endowments) of every consumer in the economy. In this section, we shall explore the implications of assuming that this cannot be done to any extent; that is, we shall postulate that individual characteristics are private and become public only if revealed by economic agents through their choices. We will then see that under very general conditions the second welfare theorem fails dramatically: the only Pareto optimal allocations that can be supported involve no transfers, that is, they are precisely the Walrasian allocations. Thus, if no personal information of any sort is available to the policy authority, then there may be a real conflict between equity and efficiency: if transfers have to be implemented we must give up Pareto optimality. The nature of this trade-off is further explored in Sections 22.B and 22.C.

We place ourselves in an exchange economy with I consumers. Each consumer i has the consumption set $\mathbb{R}_+^L$, the endowment vector $\omega_i \geq 0$, and the continuous, monotone, and strictly quasiconcave utility function $u_i(\cdot)$.

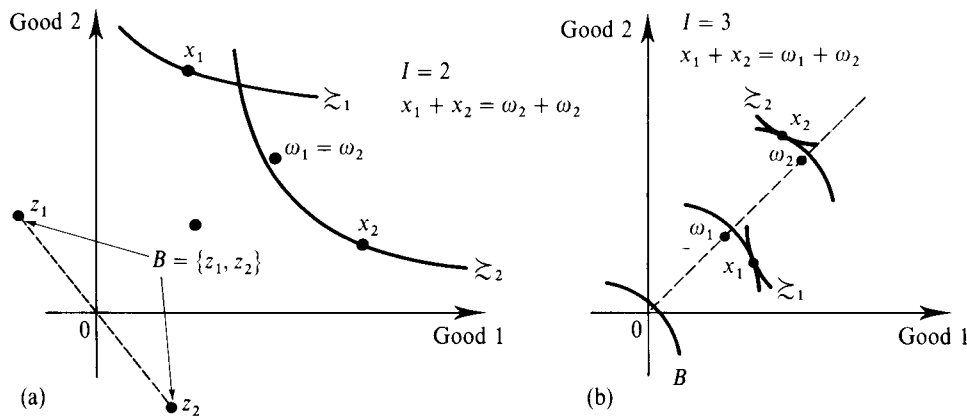


Figure 18.D.1
(a) and (b): Two self-selective allocations.

We begin by stating a restriction on feasible allocations designed to capture the possibility that the allocation is the result of a process in which every consumer maximizes utility subject to market opportunities that are identical across consumers.

Definition 18.D.1: The feasible allocation $x^* = (x_1^*, \dots, x_I^*) \in \mathbb{R}^{LI}$ is *self-selective* (or *anonymous*, or *envy-free in net trades*) if there is a set of net trades $B \subset \mathbb{R}^L$, to be called a *generalized budget set* or a *tax system*, such that, for every i , $z_i^* = x_i^* - \omega_i$ solves the problem

$$\begin{aligned} \text{Max } & u_i(z_i + \omega_i) \\ \text{s.t. } & z_i \in B, \\ & z_i + \omega_i \geq 0. \end{aligned}$$

Figures 18.D.1(a) and 18.D.1(b) present two examples of self-selective allocations.¹⁵ In the figures the preferences and endowments of the different consumers are depicted in the same orthant.

Note that if $x^* = (x_1^*, \dots, x_I^*)$ is self-selective then it is enough to take $B = \{x_1^*, \dots, x_I^*\}$. Thus, we could read Definition 18.D.1 as saying that no consumer i envies the net trade of any other individual; among all the net trades present in the economy, the consumer is happy enough with the one assigned to her. Expositionally, we have preferred to keep separate the reality of B because we have in mind the limit situation in which there is, on the one hand, a multitude of consumers whose actions are individually imperceptible and, on the other, a policy authority that has perfect statistical information (i.e., knows perfectly the *distribution* of individual characteristics), but no information at all on who is who. In such a world, a viable policy instrument is to choose a set B and let each consumer select her most preferred point in it. Because this is what an income tax schedule amounts to, we also call B a tax system. We remark that, as a policy tool, the notion of a generalized budget presumes the ability to prevent individuals from choosing several times. Hence it models the income tax, adequately, but not a commodity tax.

We now pose a question of the second welfare theorem type: Which Pareto optimal allocations can be supported by means of a common budget? That is, which feasible allocations are simultaneously Pareto optimal and self-selective? This

15. The concept of a nonenvy allocation was introduced by Foley (1967) and that of a nonenvy net-trade allocation by Schmeidler and Vind (1972). See Thomson and Varian (1985) for a survey of these notions (with an emphasis on the ethical aspects).

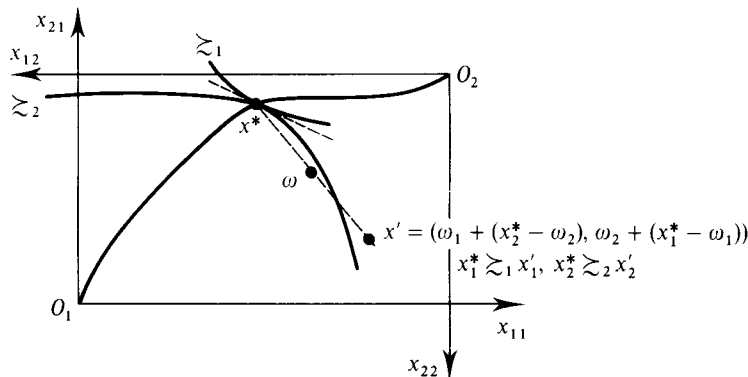


Figure 18.D.2

A Pareto optimal, self-selective allocation that is not Walrasian.

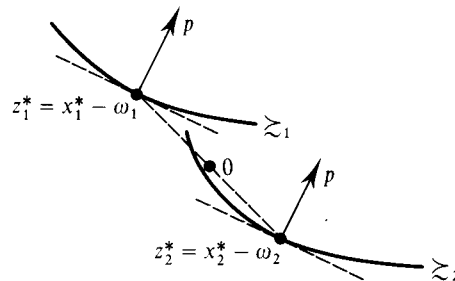


Figure 18.D.3

Another representation of the example of Figure 18.D.2.

question is both broader and more limited than the one associated with the second welfare theorem. It is broader because we allow supportability by general (nonlinear) budget sets and not only by linear hyperplanes. It is narrower because it demands that *all* consumers face the *same* budget for their net trades.

The first observation is that if $x^* = (x_1^*, \dots, x_T^*)$ is a Walrasian allocation with equilibrium price vector $p \in \mathbb{R}^L$, then the allocation is Pareto optimal because of the first welfare theorem, and self-selective because we can take $B = \{z: p \cdot z = 0\}$.

The allocation marked x^* in the Edgeworth box of Figure 18.D.2 provides an example of a self-selective allocation in the Pareto set that is not Walrasian. Figure 18.D.3 shows transparently why x^* is self-selective. In this figure we bring the origin of the consumption sets of the two consumers to their initial endowment vector so that the preferences of the two consumers are expressed over net trades. Then with $z_i^* = x_i^* - \omega_i$, $i = 1, 2$, we see that $z_1^* \succeq_1 z_2^*$ and $z_2^* \succeq_2 z_1^*$.

The previous Edgeworth box example suggests that there may be ample room for anonymous redistribution. The example, however, is special in that there are only two consumers or, more generally, in that all the consumers fall into two preference–endowment types. We will now investigate the situation in which there is a multitude of consumers who *moreover* fall into a rich variety of types. Intuitively, this should make the compatibility of Pareto optimality and transfers more difficult because it is likely that the opportunities for envy will then be many (Pareto optimality will force the net trades to vary across consumers), and therefore the freedom in constructing the generalized budget will be limited.

In Figure 18.D.4 we represent an allocation for an economy with two commodities and a continuum of types (and therefore with a continuum of consumers).¹⁶

16. Approximate versions of the following results exist for economies with a finite but large number of types.

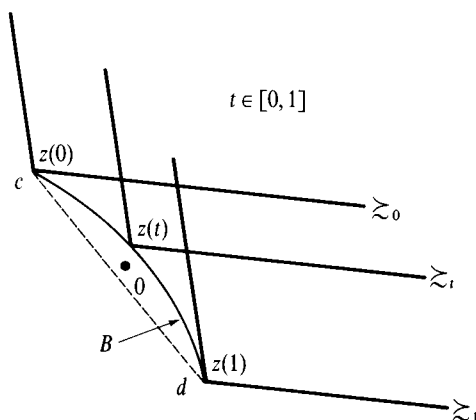


Figure 18.D.4
Pareto optimal allocation with a continuum of traders that is self-selective but not Walrasian.

Consumer types are indexed by $t \in [0, 1]$ and their preferences $\succsim_t$ depend continuously on t . For simplicity we take their endowments to be the same.¹⁷ An implication of this continuity assumption is that the set of characteristics (preferences–endowment pairs) of the consumers present in the economy cannot be split into two disconnected classes. The consumptions of the different types are distributed along the curvilinear segment cd . This allocation has the following properties:

- (i) It is not Walrasian. If it were then all the consumptions would lie in a straight line; in fact, different types end up exchanging the two goods at different ratios.
- (ii) It is self-selective. We see in Figure 18.D.4 that the consumption chosen by each consumer maximizes her utility in the generalized budget set B . Note that the frontier of any admissible budget set has to include the segment cd comprised by the consumptions actually chosen by some consumer.
- (iii) It is Pareto optimal. Indeed, the price vector $p = (1, 1)$ will make the allocation into a price equilibrium with transfers, hence a Pareto optimum.

Observe that fact (iii) depends crucially on the indifference curves of every consumer exhibiting a kink at the assigned bundle. If we tried to smooth these kinks out then, because cd is curved and the preferences change continuously, the result would be the existence of two consumers with different marginal rates of substitution at their chosen point, which is a violation of Pareto optimality (there would be room for profitable exchange among these two consumers). Only if B were rectilinear could we retain Pareto optimality, but then the allocation would be a Walrasian equilibrium. Thus, it appears that if the indifference curves are smooth at the consumption points then a Pareto optimal, self-selective allocation can be something other than a Walrasian equilibrium only if the characteristics of the consumers present in the economy can be split into disconnected classes. With this motivation we can state Proposition 18.D.1.¹⁸

17. What is important is that they change continuously with t .

18. For results of this type, see, for example, Varian (1976) or Champsaur and Laroque (1981).

Proposition 18.D.1: Suppose we have an exchange economy with a continuum of consumer types. Assume:

- (i) The preferences of all consumers are representable by differentiable utility functions.
- (ii) The set of characteristics of consumers present in the economy¹⁹ cannot be split into two disconnected classes. Formally, if $(u(\cdot), \omega), (u'(\cdot), \omega')$ are two preference–endowment pairs present in the economy then there is a continuous function $(u(\cdot; t), \omega(t))$ of $t \in [0, 1]$ such that

$$(u(\cdot; 0), \omega(0)) = (u(\cdot), \omega), (u(\cdot; 1), \omega(1)) = (u'(\cdot), \omega'),$$

and $(u(\cdot; t), \omega(t))$ is present in the economy for every t .

Then any allocation $x^* = \{x_i^*\}_{i \in I}$ that is Pareto optimal, self-selective, and interior (i.e., $x_i^* \gg 0$ for all i) must be a Walrasian equilibrium allocation. Here I is an infinite set of names.

Proof: The proof is far from rigorous. It is also limited to $L = 2$.

Let $p = (p_1, p_2)$ be the price vector supporting x^* as a Pareto optimal allocation. Because of differentiability of the utility functions and interiority of the allocation, the relative prices p_1/p_2 are uniquely determined. We want to show that $p \cdot (x_i^* - \omega_i) = 0$ for all i .

The first observation is that at x^* the equal-treatment property holds: if $(u_i(\cdot), \omega_i) = (u_k(\cdot), \omega_k)$ then $x_i^* = x_k^*$. Indeed, neither i envies k nor k envies i . Hence x_i^* and x_k^* must lie in the same indifference curve of the common preference relation of i and k . By the strict convexity of preferences, the price vector p can support only one point in this indifference curve. Hence x_i^* and x_k^* must be equal.

If the set of net trades present in the economy consists of a single point, then this point has to be the vector 0 (otherwise the aggregate of the net trades could not be zero) and the result follows.

Suppose, therefore, that there are at least two different net-trade vectors present in the economy, z_0 and z_1 . In Figure 18.D.5, we represent them as well as the net trades $z(t)$ of all consumers captured by the continuous parametrization given by assumption (ii), where $t = 0, 1$ correspond, respectively, to consumers underlying z_0 and z_1 .

A key fact is that $z(t)$ is continuous as a function of t . This is intuitive. We have already seen that the equal-treatment property holds: Identical individuals are treated identically. Technicalities aside, the logic of the continuity of $z(t)$ is the same: If envy is to be prevented, then similar individuals must be treated similarly.

Thus as we go from $t = 0$ to $t = 1$, the net trades are moving continuously from z_0 to z_1 . Hence the frontier of any generalized budget set B must actually connect z_0 to z_1 . If so, then either this frontier is a straight segment with normal p between these two points [in which case $p \cdot (z_0 - z_1) = 0$], or somewhere between them there is one point $(z(t'))$ in Figure 18.D.5 where the “slope” of the frontier of B , and therefore the *MRS* of the consumer choosing this point, is different from p_1/p_2 [in which case p would not be a supporting price vector].

We conclude that the portion M of the frontier of B containing all the net trades present in the economy is a nontrivial straight segment with normal p (hence, a convex set). Since the net trades add up to zero, we must have $0 \in M$. Thus, $p \cdot z = p \cdot (z - 0) = 0$ for every $z \in M$. In particular, $p \cdot (x_i^* - \omega_i) = 0$ for every i . See Figure 18.D.6. ■

19. The expression “characteristics of consumers present in the economy” means technically “contained in the support of the distribution of characteristics induced by the population of consumers.”

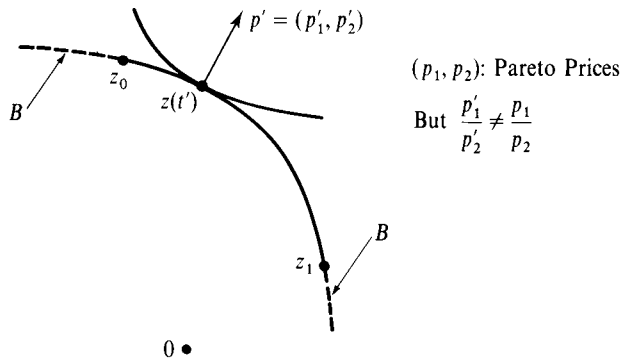


Figure 18.D.5. (left)
If the net trade frontier is not flat, then the allocation is not Pareto optimal (assuming self-selectivity).

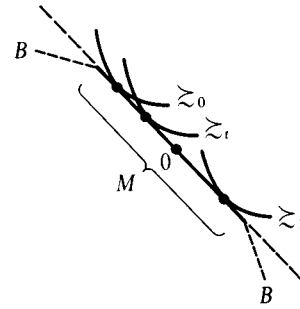


Figure 18.D.6 (right)
A Pareto optimal, self-selective allocation that is Walrasian.

18.E Equilibrium and the Marginal Productivity Principle

In this section, we investigate the extent to which Walrasian equilibria can be characterized by the idea that individuals get exactly what they contribute to the economic welfare of society (at the margin). We will see that, once again, the assumption of a large number of consumers is crucial to this characterization. For an extensive analytical treatment of this topic we refer to the seminal contribution of Ostroy (1980).

To remain as simple as possible, we restrict ourselves to the case of quasilinear exchange economies. The L th good is the numeraire.

Suppose that our economy has H types. The concave, differentiable, strictly increasing utility function of type h is

$$u_h(x_h) = \psi_h(x_{1h}, \dots, x_{L-1,h}) + x_{Lh}.$$

It is defined on $\mathbb{R}_+^{L-1} \times \mathbb{R}$. We take ψ_h to be strictly concave. The initial endowment vector of type h is $\omega_h \geq 0$.

An economy is defined by a profile $(I_1, \dots, I_H)$ of consumers of the different types, for a grand total of $I = \sum_h I_h$. For any economy $(I_1, \dots, I_H)$ we define the maximal amount of “social utility” that can be generated, as in Section 10.D.²⁰

$$\begin{aligned} v(I_1, \dots, I_H) = \text{Max} \quad & I_1 u_1(x_1) + \dots + I_H u_H(x_H) \\ \text{s.t.} \quad & \text{(i) } I_1 x_1 + \dots + I_H x_H \leq I_1 \omega_1 + \dots + I_H \omega_H, \\ & \text{(ii) } x_{\ell h} \geq 0 \text{ for all } \ell \leq L-1 \text{ and } h. \end{aligned} \quad (18.E.1)$$

The function $v(I_1, \dots, I_H)$ is homogeneous of degree one in its arguments: $v(rI_1, \dots, rI_H) = rv(I_1, \dots, I_H)$ for all r . In particular,

$$v(I_1/I, \dots, I_H/I) = \frac{1}{I} v(I_1, \dots, I_H).$$

That is, the per-capita social utility only depends on the type-composition and not on the size of the economy. Because of this we can extend the analysis to a situation with a continuum of consumers by defining $v(\mu_1, \dots, \mu_H)$ for any nonnegative vector

20. Because utility functions are concave the maximum utility can be reached while treating consumers of the same type equally.

$\mu = (\mu_1, \dots, \mu_H) \in \mathbb{R}_+^H$ of masses of the different types. Precisely,

$$v(\mu_1, \dots, \mu_H) = \text{Max} \quad \mu_1 u_1(x_1) + \dots + \mu_H u_H(x_H) \quad (18.E.2)$$

$$\text{s.t. (i) } \mu_1 x_1 + \dots + \mu_H x_H \leq \mu_1 \omega_1 + \dots + \mu_H \omega_H,$$

$$\text{(ii) } x_{\ell h} \geq 0 \text{ for all } \ell \leq L-1 \text{ and } h.$$

If we have a sequence of finite economies $(I_1^n, \dots, I_H^n)$ such that $I^n = \sum_h I_h^n \rightarrow \infty$ and $(1/I^n)I_h^n \rightarrow \mu_h$ for every h , then we can properly regard $(\mu_1, \dots, \mu_H)$ as the continuum limit of the sequence of increasingly large finite economies.

Exercise 18.E.1: Show that the function $v(\cdot): \mathbb{R}_+^H \rightarrow \mathbb{R}$ is concave and homogeneous of degree one.

The function $v(\cdot)$ is a sort of production function whose output is social utility and whose inputs are the individual consumers themselves. Further, in the limit, every individual of type h becomes an input of infinitesimal size. For the time being, we concentrate our discussion on the continuum limit. We assume also that $v(\cdot)$ is differentiable.²¹

Definition 18.E.1: Given a continuum population $\mu = (\mu_1, \dots, \mu_H) \in \mathbb{R}_+^H$ a feasible allocation²² $(x_1^*, \dots, x_H^*)$ is a *marginal product*, or *no-surplus*, allocation if

$$u_h(x_h^*) = \frac{\partial v(\mu)}{\partial \mu_h} \quad \text{for all } h. \quad (18.E.3)$$

In words: at a no-surplus allocation everyone is getting exactly what she contributes at the margin.

Proposition 18.E.1: For any *continuum* population $\bar{\mu} = (\bar{\mu}_1, \dots, \bar{\mu}_H) \gg 0$ a feasible allocation $(x_1^*, \dots, x_H^*) \gg 0$ is a marginal product allocation if and only if it is a Walrasian equilibrium allocation.

Proof: If $x^* = (x_1^*, \dots, x_H^*)$ is a marginal product allocation then, using Euler's formula (see Section M.B of the Mathematical Appendix), we have

$$v(\bar{\mu}) = \sum_h \bar{\mu}_h \frac{\partial v(\bar{\mu})}{\partial \mu_h} = \sum_h \bar{\mu}_h u_h(x_h^*).$$

Hence, x^* solves problem (18.E.2) for $\mu = \bar{\mu}$.

Suppose now that $x^* = (x_1^*, \dots, x_H^*)$ is a feasible allocation that gives rise to social utility $v(\bar{\mu})$; that is, it constitutes a solution to problem (18.E.2) for $\mu = \bar{\mu}$. Denote by p_ℓ , $\ell = 1, \dots, L$, the values of the multipliers of the first-order conditions associated with the constraints $\sum_h \bar{\mu}_h (x_{\ell h} - \omega_{\ell h}) \leq 0$, $\ell = 1, \dots, L$, in the optimization problem (18.E.2); see Section M.K of the Mathematical Appendix. By the quasilinear form of $u_h(\cdot)$ we have

$$p_L = 1 \quad \text{and} \quad p_\ell = \nabla_\ell \psi_h(x_{1h}^*, \dots, x_{L-1,h}^*) \quad (18.E.4)$$

for all $\ell \leq L-1$ and all $1 \leq h \leq H$.

21. This could be derived from more primitive assumptions.

22. We assume that consumers of the same type are treated equally. Feasibility means therefore that $\sum_h \mu_h x_h^* \leq \sum_h \mu_h \omega_h$.

It follows from (18.E.4) that the vector of multipliers $p = (p_1, \dots, p_L)$ is the vector of Walrasian equilibrium prices of this quasilinear economy (recall the analysis of Section 10.D). In addition, by the envelope theorem (see Section M.L of the Mathematical Appendix), applied to problem (18.E.2), we have (Exercise 18.E.2):

$$\frac{\partial v(\bar{\mu})}{\partial \mu_h} = u_h(x_h^*) + p \cdot (\omega_h - x_h^*). \quad (18.E.5)$$

Therefore, we conclude that x^* is Walrasian if and only if x^* solves problem (18.E.2) for $\mu = \bar{\mu}$ and (18.E.3) is satisfied, that is, if and only if x^* is a marginal product allocation. ■

Expression (18.E.5) is intuitive. The left-hand side measures how much the maximum sum of utilities increases if we add one extra individual of type h . The right-hand side tells us that there are two effects. On the one hand, the extra consumer of type h receives from the rest of the economy the consumption bundle x_h^* , and so she directly adds her utility $u_h(x_h^*)$ to the social utility sum. On the other, while receiving x_h^* , she contributes her endowment vector ω_h . Hence the net change for the rest of the economy is $\omega_h - x_h^*$. How much is this worth to the rest of the economy? The vector of social shadow prices is precisely $p = (p_1, \dots, p_L)$, and so the total change for the rest of the economy comes to $p \cdot (\omega_h - x_h^*)$. Note that the Walrasian allocations are thus characterized by this second effect being null: the utility of the consumer equals her entire marginal contribution to social utility.

In Exercise 18.E.4 you are asked to verify that the smoothness assumption on utility functions is essential to the validity of Proposition 18.E.1.

Let us now consider a finite economy $(I_1, \dots, I_H) \gg 0$. We can define the marginal contribution of an individual of type h as

$$\Delta_h v(I_1, \dots, I_H) = v(I_1, \dots, I_h, \dots, I_H) - v(I_1, \dots, I_h - 1, \dots, I_H).$$

Typically, there does not exist a feasible allocation $(x_1^*, \dots, x_H^*)$ with $u_h(x_h^*) = \Delta_h v(I_1, \dots, I_H)$ for all h . To see this, note that by the concavity of $v(\cdot)$ we have $\Delta_h v \geq \partial v / \partial \mu_h$ [both expressions evaluated at $(I_1, \dots, I_H)$]. Except for degenerate cases, this inequality will be strict. Moreover, $\sum_h I_h (\partial v / \partial \mu_h) = v(I_1, \dots, I_H)$ by Euler's formula (see Section M.B of the Mathematical Appendix), and thus we conclude that $\sum_h I_h (\Delta_h v) > v(I_1, \dots, I_H)$; that is, it is impossible to give to each consumer the full extent of her marginal contribution while maintaining feasibility. In contrast with the continuum case, individuals are not now of negligible size: their whole contribution is not entirely at the margin. In particular, you should note that in a finite economy the Walrasian allocation is typically *not* a marginal product allocation. It follows from expression (18.E.5) that an allocation $(x_1^*, \dots, x_H^*)$ that solves problem (18.E.2) for $(\mu_1, \dots, \mu_H) = (I_1, \dots, I_H)$ is a Walrasian equilibrium allocation if and only if

$$u_h(x_h^*) = \frac{\partial v}{\partial \mu_h}(I_1, \dots, I_H).$$

But we have just argued that normally $\Delta_h v(I_1, \dots, I_H) > \partial v(I_1, \dots, I_H) / \partial \mu_h$. In words: At the Walrasian equilibrium consumers are compensated according to prices determined by the marginal unit of their endowments. But they lose the extra social surplus provided by the inframarginal units. This is yet another indication that the concept of Walrasian equilibrium stands on firmer ground in large economies.

We have just seen that in the context of economies with finitely many consumers it is not possible to feasibly distribute the gains of trade while adhering literally to the marginal productivity principle. The cooperative theory of games provides a possibility for a sort of reconciliation between feasibility and the marginal productivity principle. It is known as the *Shapley value*. In Appendix A, devoted to cooperative game theory, we offer a detailed presentation of this solution concept.

For an economy with profile $(I_1, \dots, I_H)$ the Shapley value is a certain utility vector $(Sh_1, \dots, Sh_H) \in \mathbb{R}^H$ that satisfies $\sum_h I_h Sh_h = v(I_1, \dots, I_H)$. For every type h , the utility Sh_h can be viewed as an *average of marginal utilities* $\Delta_h v(I'_1, \dots, I'_H)$. The average is taken over profiles $(I'_1, \dots, I'_H) \leq (I_1, \dots, I_H)$, where the probability weight given to $(I'_1, \dots, I'_H)$ equals $1/I$, interpreted as the probability assigned to sample size $I'_1 + \dots + I'_H$, times the probability of getting the profile $(I'_1, \dots, I'_H)$ when independently sampling $I'_1 + \dots + I'_H$ consumers out of the original population with I consumers and profile $(I_1, \dots, I_H)$. See Appendix A for more on this formula.

An allocation that yields the Shapley value (let us call it a *Shapley allocation*) is not related in any particular way to the Walrasian equilibrium allocation (or for that matter to the core). Except by chance, they will be different allocations. Yet, remarkably, we also have a convergence of these concepts in economies with many consumers: the Walrasian and the Shapley allocations are then close to each other. This result is known as the *value equivalence theorem*. A rigorous proof of this theorem is too advanced to be given here [see Aumann (1975) and his references], but the basic intuition is relatively straightforward.

There are two key facts. First, if the entries of $(I'_1, \dots, I'_H)$ are large, then subtracting a consumer of type h amounts to very little, and so

$$\Delta_h v(I'_1, \dots, I'_H) \approx \partial v(I'_1, \dots, I'_H) / \partial \mu_h.$$

Second, if the entries of $(I_1, \dots, I_H)$ are large then, by the law of large numbers, most profiles $(I'_1, \dots, I'_H)$ constitute a good sample of $(I_1, \dots, I_H)$ and are therefore almost proportional to $(I_1, \dots, I_H)$.

Using the homogeneity of degree one of $v(\cdot)$ (hence the homogeneity of degree zero of $\partial v / \partial \mu_h$), the combination of the previous two facts implies

$$\Delta_h v(I'_1, \dots, I'_H) \approx \frac{\partial v(I'_1, \dots, I'_H)}{\partial \mu_h} \approx \frac{\partial v(I_1, \dots, I_H)}{\partial \mu_h}$$

for most $(I'_1, \dots, I'_H)$. Therefore, $Sh_h \approx \partial v(I_1, \dots, I_H) / \partial \mu_h$, which is the utility payoff of type h at the Walrasian equilibrium allocation of the economy $(I_1, \dots, I_H)$.

APPENDIX A: COOPERATIVE GAME THEORY

In this appendix, we offer a brief introduction to the *cooperative theory of games*. For more extensive recent accounts see Moulin (1988), Myerson (1991), or Osborne and Rubinstein (1994).²³

In Chapter 7, we presented the normal and the extensive forms of a game. The starting point for the cooperative theory is a classical third description: that of a

23. The text by Owen (1982), although not so recent, is nevertheless strong in its coverage of cooperative theory. Another useful reference is Shubik (1984), which is encyclopedic in spirit and contains a wealth of information.

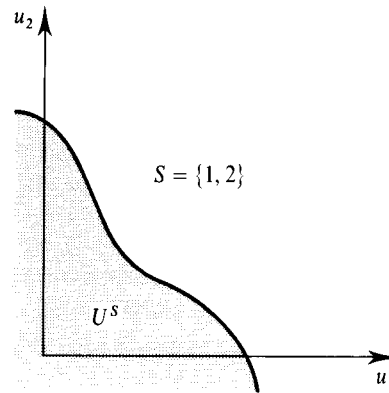
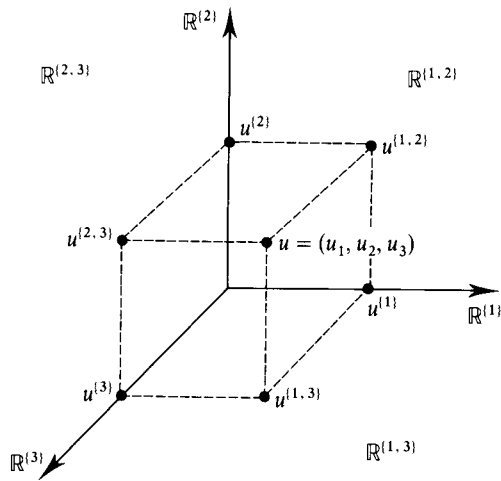


Figure 18.AA.1 (left)
Utility outcome $u \in \mathbb{R}^3$
and its projections.

Figure 18.AA.2 (right)
A utility possibility set
for $S = \{1, 2\}$.

game in characteristic form. The characteristic form is meant to be a summary of the payoffs available to each group of players in a context where binding commitments among the players of the group are feasible. Although, in principle, it should be possible to derive the characteristic form from the normal or the extensive forms, the viewpoint of cooperative game theory is that it is often analytically desirable to avoid detail and to go as directly as possible to a summary description of the strategic position of the different groups of players.²⁴

After defining the characteristic form, we will discuss two of the main solution concepts of cooperative game theory: the *core* and the *Shapley value*.

The set of players is denoted $I = \{1, \dots, I\}$. We abuse notation slightly by using the same symbol to denote the set and its cardinality. Nonempty subsets $S, T \subset I$ are called *coalitions*.

An *outcome* is a list of utilities $u = (u_1, \dots, u_I) \in \mathbb{R}^I$. Given $u = (u_1, \dots, u_I)$, the relevant coordinates to a coalition S are $u^S = (u_i)_{i \in S}$. Mathematically, u^S is the restriction (or projection) of $u \in \mathbb{R}^I$ to the coordinates corresponding to S . We can therefore view u^S as a member of the Euclidean space $\mathbb{R}^S$ spanned by these coordinates. Figure 18.AA.1 shows how outcomes for three players are evaluated by all six proper subsets: $S = \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}$, and $\{2, 3\}$.

Definition 18.AA.1: A nonempty, closed set $U^S \subset \mathbb{R}^S$ is a *utility possibility set* for $S \subset I$ if it is *comprehensive*:

$$u^S \in U^S \quad \text{and} \quad u'^S \leq u^S \text{ implies } u'^S \in U^S.$$

See Figure 18.AA.2 for an illustration.²⁵

24. Nevertheless, we note that there is a school of thought within game theory of the opinion that the condensation of information that a characteristic form represents may not do justice to the strategic complexities inherent in the making of binding commitments. Despite the cogency of this position, the analytical power of games in characteristic forms for the study of normative issues in economics has been amply demonstrated. This is more than enough reason to welcome the parsimony it brings to the analysis.

25. Note that, as we did in Section 16.E, we build free disposability of utility into the definition of a utility possibility set.

Definition 18.AA.2: A game in characteristic form (I, V) is a set of players I and a rule $V(\cdot)$ that associates to every coalition $S \subset I$ a utility possibility set $V(S) \subset \mathbb{R}^S$.

The elements of $V(S)$ are to be interpreted as the payoffs the players in S can achieve by themselves if they jointly commit to a certain course of action. It is important to observe the expression “can achieve” is not free of subtlety. This is because the course of action undertaken by the members of $I \setminus S$ will typically affect the payoffs of the members of S . In applications, therefore, one should be explicit as to how $V(S)$ is constructed.

Example 18.AA.1: Economies. Consider an economy with I consumers having continuous, increasing, concave utility functions $u_i: \mathbb{R}_+^L \rightarrow \mathbb{R}$ and endowments $\omega_i \geq 0$. There is also a publicly available convex, constant returns technology $Y \subset \mathbb{R}^L$. We can then define a game in characteristic form by letting

$$V(S) = \left\{ (u_i(x_i))_{i \in S} : \sum_{i \in S} x_i = \sum_{i \in S} \omega_i + y, y \in Y \right\} - \mathbb{R}_+^S.$$

That is, $V(S)$ is the set of payoffs that the consumers in coalition S can achieve by trading among themselves and using the technology Y . Every set $V(S)$ is convex (recall Exercise 16.E.2). Figure 18.AA.3 depicts these sets for the case $I = 3$. ■

Example 18.AA.2: Majority Voting. Consider a three-player situation in which any two out of the three players can form a majority and select among a set of social alternatives A . If $a \in A$ is selected, the payoffs are $u_i(a) \geq 0$, $i = 1, 2, 3$. In addition, any player i has the right to unilaterally withdraw from the group and get a payoff of zero.

Then we can define a game in characteristic form (I, V) as

$$V(I) = \{(u_1(a), u_2(a), u_3(a)) : a \in A\} - \mathbb{R}_+^I.$$

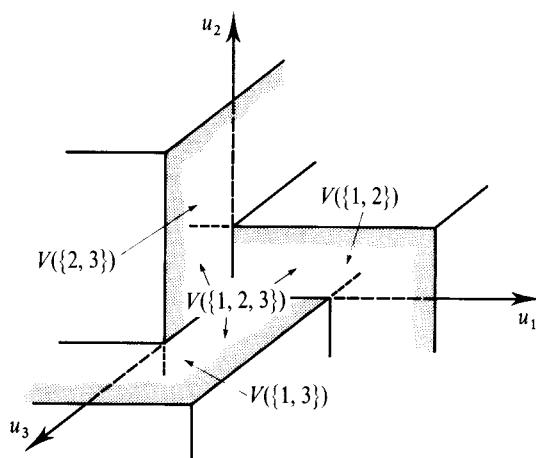
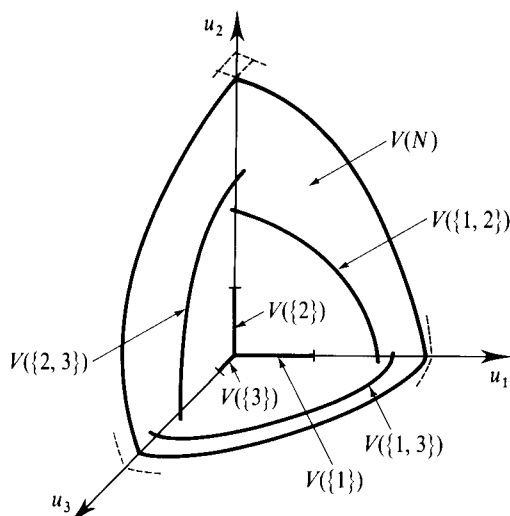
$$V(\{i, h\}) = \{(u_i(a), u_h(a)) : a \in A\} - \mathbb{R}_+^{(i, h)} \text{ for all distinct pairs } \{i, h\}.$$

$$V(\{i\}) = -\mathbb{R}_+^{(i)}.$$

Figure 18.AA.4 shows this characteristic form for a case with three alternatives, $A = \{a_1, a_2, a_3\}$. In the figure we suppose that the three decisions yield, respectively,

Figure 18.AA.3 (left)
A family of utility possibility sets.

Figure 18.AA.4 (right)
The utility possibility sets for the majority voting in Example 18.AA.2.



the utility vectors $(2, 1, 0)$, $(1, 0, 2)$, $(0, 2, 1)$. Note that $V(\cdot)$ need not be convex when, as here, decisions are discrete and there are no possibilities for either randomization or any form of side transfers. ■

Definition 18.AA.3: A game in characteristic form (I, V) is *superadditive* if for any coalitions $S, T \subset I$ that are disjoint (i.e., such that $S \cap T = \emptyset$), we have

$$\text{If } u^S \in V(S) \text{ and } u^T \in V(T), \text{ then } (u^S, u^T) \in V(S \cup T).$$

Superadditivity means that coalitions S and T are able to do at least as well acting together as they could do acting separately. It is an assumption we will commonly make (it is satisfied by Examples 18.AA.1 and 18.AA.2). If one of the possibilities open to the union of two disjoint coalitions is to agree to act *as if* they were still separated coalitions, then superadditivity should hold.

It has been a constant theme of this book that often the analysis becomes much simpler when individual utility functions are quasilinear, that is, when there is a commodity (“numeraire”) that can be used to effect unit-per-unit transfers of utility across economic agents. The same is true in the theory of cooperative games. Its history is, in fact, replete with instances of concepts first formulated for the transferable utility case that have later been extended to the general setting without an essential loss of intuition and analytical power.

For a situation described by a game in characteristic form, what the quasilinearity, or transferable utility, hypothesis amounts to is the assumption that the sets $V(S)$ are half-spaces (as they were, for example, in Section 10.D); that is, they are sets whose boundaries are hyperplanes in $\mathbb{R}^S$. Moreover, by choosing the units of utility, we can take the hyperplanes defining $V(S)$ to have normals $(1, \dots, 1) \in \mathbb{R}^S$.²⁶ Thus, the sets $V(S)$ will now have the form

$$V(S) = \left\{ u^S \in \mathbb{R}^S : \sum_{i \in S} u_i^S \leq v(S) \right\}$$

for some $v(S) \in \mathbb{R}$. In other words, we can view coalition S as choosing a joint action so as to maximize the total utility, denoted $v(S)$, which then can be allocated to the members of S in any desired manner through transfers of the numeraire. Figure 18.AA.5 depicts the sets $V(S)$ for the case $I = 3$.

The number $v(S)$ is called the *worth* of coalition S . Since the numbers $v(S)$, $S \subset I$, constitute a complete description of (I, V) we provide Definition 18.AA.4.

Definition 18.AA.4: A *transferable utility game in characteristic form*, (or *TU-game*), is defined by (I, v) , where I is a set of players and $v(\cdot)$ is a function, called the *characteristic function*, that assigns to every nonempty coalition $S \subset I$ a number $v(S)$ called the *worth of S*.

Example 18.AA.3: TU Majority Voting. Suppose that in Example 18.AA.2 (with the values of Figure 18.AA.4) we add the possibility that utility be freely transferable across players (there may be a numeraire commodity with respect to which

26. This choice of units of utility is legitimate because all the solutions to be considered are invariant to normalizations of units. See Chapter 21 for more on this point.

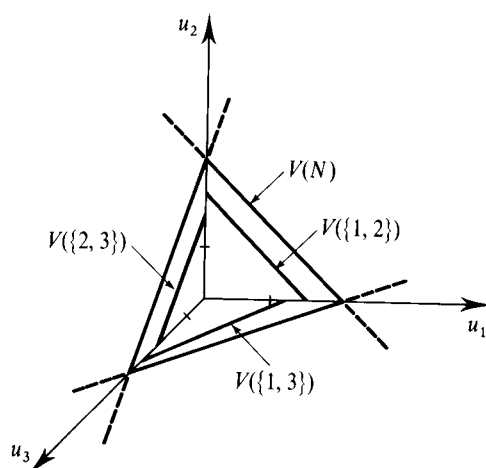


Figure 18.AA.5

Utility possibility sets for a transferable utility game.

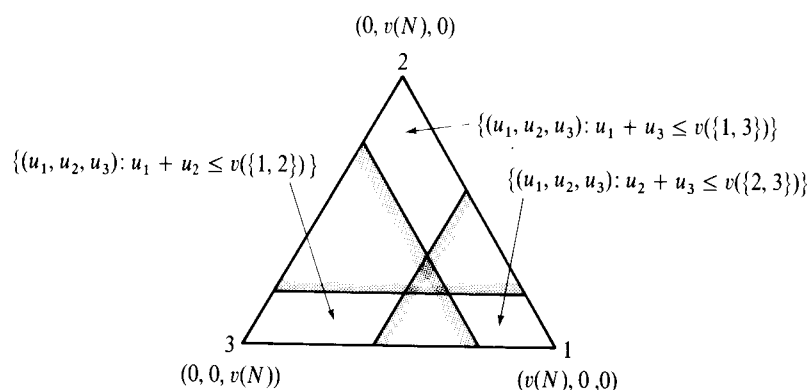


Figure 18.AA.6

Using a simplex to represent a three-player TU-game with utilities normalized so that $V(\{i\}) = 0$.

preferences are quasilinear). Then the characteristic function is

$$v(I) = 3, \quad v(\{1, 2\}) = v(\{1, 3\}) = v(\{2, 3\}) = 3 \quad \text{and} \quad v(\{i\}) = 0, \quad i = 1, 2, 3. \quad \blacksquare$$

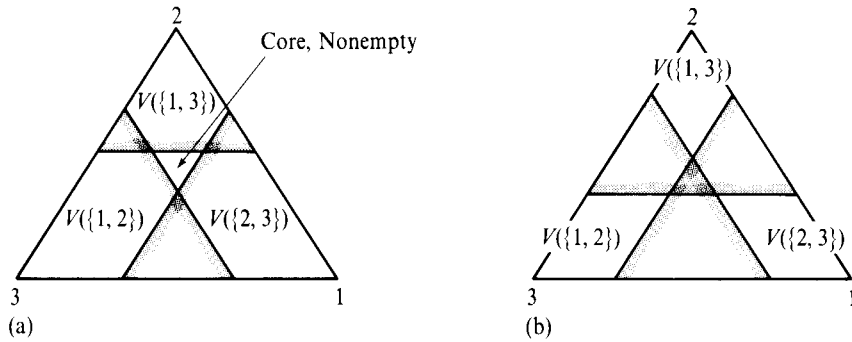
All the developments in this appendix will be invariant to changes in the origins of individual utilities; thus we can fix these arbitrarily. The usual convention is to put $v(\{i\}) = 0$ for every i .

In Figure 18.AA.6 we represent a diagrammatic device for the case of three-player games that is particularly useful. Instead of working in three dimensions, we look at the two-dimensional simplex that exhibits all possible divisions of $v(I)$ subject to the condition that $u_i \geq 0$ for all i [which means, in the normalization just discussed, that $u_i \geq v(\{i\})$]. The other sets in the diagram represent, for every two-person coalition $\{i, h\}$, the utility combinations in the simplex satisfying $u_i + u_h \leq v(\{i, h\})$.

We now turn to a presentation of two well-known solution concepts for cooperative games: the *core* and the *Shapley value*.

The Core

The first solution concept we review is the *core*: the set of feasible utility outcomes with the property that no coalition could on its own improve the payoffs of all its members. An empty core is indicative of competitive instability in the situation being modeled. If the core is nonempty and small, then we could say that coalitional

**Figure 18.AA.7**

(a) A TU-game with a nonempty core.
 (b) A TU game with an empty core.

competition by itself brings about a sharply determined outcome. If it is nonempty and large, then coalitional competition alone does not narrow down the possible outcomes very much.

Definition 18.AA.5: Given a game in characteristic form (I, V) , the utility outcome $u \in \mathbb{R}^I$ is *blocked*, or *improved upon*, by a coalition $S \subset I$ if there exists $u'^S \in V(S)$ such that $u'_i < u_i$ for all $i \in S$.

If the game is a TU game (I, v) then the outcome $u = (u_1, \dots, u_I)$ is blocked by S if and only if $\sum_{i \in S} u_i < v(S)$.

Definition 18.AA.6: A utility outcome $u = (u_1, \dots, u_I)$ that is feasible for the grand coalition [i.e., $u \in V(I)$] is in the *core* of the characteristic form game (I, V) if there is no coalition S that blocks u .

In TU games the core is the set of utility vectors $u = (u_1, \dots, u_I)$ satisfying the linear inequalities

$$\sum_{i \in S} u_i \geq v(S) \text{ for all } S \subset I \quad \text{and} \quad \sum_{i \in I} u_i \leq v(I).$$

Figure 18.AA.7(a) depicts a three-player game with nonempty core. In contrast, in Figure 18.AA.7(b) the core is empty. See Exercise 18.AA.1 for a set of necessary and sufficient conditions in the TU case for the nonemptiness of the core.

Exercise 18.AA.2: Show that any TU game with a nonempty core must satisfy: For any two coalitions $S, T \subset I$ such that $S \cap T = \emptyset$ and $S \cup T = I$, we have $v(S) + v(T) \leq v(I)$.

Example 18.AA.4: Majority Voting, Once Again. For the majority voting games described in Examples 18.AA.2 and 18.AA.3, the core is empty. In the latter, which is a TU game, this is clear enough: if $u_1 + u_2 + u_3 = 3$ then $u_i + u_h < 3$ for some i, h . Hence the coalition $\{i, h\}$ will block. For the former (nontransferable utility) game, note that the outcomes $(2, 1, 0)$, $(1, 0, 2)$, $(0, 2, 1)$ are blocked, respectively, by the coalitions $\{2, 3\}$ using a_3 , $\{1, 2\}$ using a_1 , and $\{1, 3\}$ using a_2 . These examples constitute instances of the so called *Condorcet paradox* (which we have already encountered in Section 1.B and will see again in Section 21.C). They are illustrative of an inherent instability of majority voting. ■

Example 18.AA.5: Economies, Again. The economic example in Example 18.AA.1 was extensively studied in Section 18.B. Note that the concept of the core discussed

there is identical with the concept considered here for games in characteristic form.²⁷ We conclude, therefore that if a Walrasian equilibrium exists then the core is nonempty. ■

Example 18.AA.6: Single-Input, Increasing Returns Production Function. Consider a one-input, one-output world in which there is a publicly available technology $f(z)$ which is continuous and satisfies $f(0) = 0$. There are I players. Each player i cares only about the consumption of the output good and owns an amount ω_i of input. Assuming that utility is transferable, we can define a TU characteristic function by $v(S) = f(\sum_{i \in S} \omega_i)$. The core of this game will be nonempty whenever the technology exhibits nondecreasing returns to scale, that is, whenever average product $f(z)/z$ is *nondecreasing*. [In particular, if $f(\cdot)$ is convex, that is, if we have a nondecreasing marginal product, then $f(z)/z$ is nondecreasing.] To verify this, suppose that we distribute the product proportionally:

$$u_h = \frac{\omega_h}{\sum_{i \in I} \omega_i} f\left(\sum_{i \in I} \omega_i\right)$$

for every $h \in I$. Then, for any $S \subset I$ we have

$$\sum_{h \in S} u_h = \frac{\sum_{h \in S} \omega_h}{\sum_{i \in I} \omega_i} f\left(\sum_{i \in I} \omega_i\right) \geq f\left(\sum_{h \in S} \omega_h\right) = v(S),$$

where the inequality follows because average product is nondecreasing. We conclude that this proportional distribution of output belongs to the core. In Exercise 18.AA.3 you are asked to show that if average product is constant then the proportional allocation is the only allocation in the core. It is also intuitively clear that the more pronounced is the degree of increasing returns, the more difficult it will be for proper subgroups to do better by their own means (they will have relatively low average product) and therefore the more we could depart from the proportional allocation while remaining in the core. Hence, for this sort of one-dimensional distribution problem, the larger the degree of increasing returns, the larger the core will be.²⁸ ■

The Shapley Value

The core tries to capture how the possible outcomes of a game may be shaped by coalitional competitive forces. It is the simplest solution concept in what could be called the *descriptive* side of cooperative game theory. We shall now investigate a solution concept, the *value*, whose motivation is normative. It attempts to describe

27. The connection of the solution concept proposed by Edgeworth (1881) with the modern game-theoretic notion of the core was made in Shubik (1959).

28. On the other hand, if $f(\cdot)$ exhibits strictly decreasing returns then it follows from Exercise 18.AA.2 that the core is empty [indeed, $v(S) + v(T) > v(I)$ for any partition of I into two coalitions S, T].

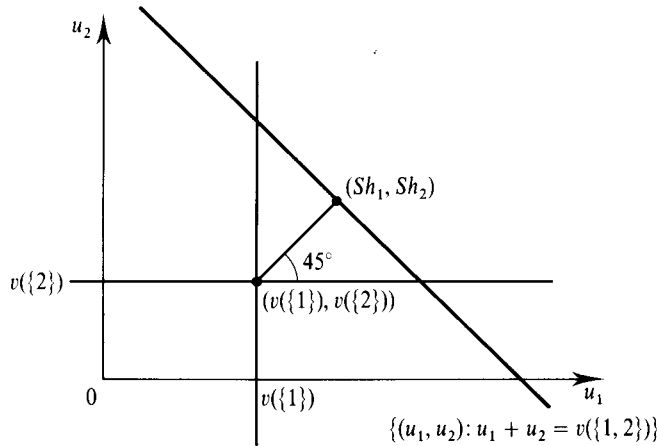


Figure 18.AA.8
Egalitarian division for two-player games.

a reasonable, or “fair,” way to divide the gains from cooperation, *taking as a given* the strategic realities captured by the characteristic form.²⁹

We study only the TU case, for which the theory is particularly simple and well established. The central concept is then a certain solution called the *Shapley value*.³⁰

Suppose that individual utilities are measured in dollars and that, so to speak, society has decided that dollars of utility of different participants are of comparable social worth. The criterion of fairness to which value theory adheres is *egalitarianism*: the aim is to distribute the gains from trade equally.

To see what the egalitarian principle could mean in the current TU context let us begin with a two-player game $(I, v) = (\{1, 2\}, v)$. Then the gains (or losses, if super-additivity fails) from cooperation are

$$v(I) - v(\{1\}) - v(\{2\}).$$

Therefore, the obvious egalitarian solution, which we denote $(Sh_1(I, v), Sh_2(I, v))$, is (see Figure 18.AA.8)

$$Sh_i(I, v) = v(\{i\}) + \frac{1}{2}(v(I) - v(\{1\}) - v(\{2\})), \quad i = 1, 2. \quad (18.AA.1)$$

How should we define the egalitarian solution $(Sh_1(I, v), \dots, Sh_I(I, v))$ for an arbitrary TU-game (I, v) ? We have already solved the problem for two-player games. It is suggestive to rewrite expression (18.AA.1) as

$$\begin{aligned} Sh_1(I, v) - Sh_1(\{1\}, v) &= Sh_2(I, v) - Sh_2(\{2\}, v), \\ Sh_1(I, v) + Sh_2(I, v) &= v(I), \end{aligned}$$

where we put $Sh_i(\{i\}, v) = v(\{i\})$. In words, this says that utility differences are preserved: *What player 1 gets out of the presence of player 2 is the same as player 2 gets out of the presence of player 1*. This points to a recursive definition: Given $S \subset I$, denote by (S, v) the TU-game obtained by restricting $v(\cdot)$ to the subsets of S [this is called a *subgame* of (I, v)]. Then we could say that a family of numbers $\{Sh_i(S, v)\}_{S \subset I, i \in S}$ constitutes an *egalitarian solution* if, for every subgame (S, v) and

29. Thus, the redistributive fairness considerations that we will discuss in Sections 22.B and 22.C, based on notions of absolute justice, are alien to the value.

30. It is named after L. Shapley, who proposed it in his Ph.D. dissertation at Princeton (in 1953).

players $i, h \in S$, utility differences are preserved in a manner similar to the two-player case:

$$Sh_i(S, v) - Sh_i(S \setminus \{h\}, v) = Sh_h(S, v) - Sh_h(S \setminus \{i\}, v) \quad \text{for all } S \subset I, i, h \in S, \quad (18.AA.2)$$

$$\sum_{i \in S} Sh_i(S, v) = v(S) \quad \text{for all } S \subset I,$$

Expressions (18.AA.2) determine the numbers $Sh_i(S, v)$, $i \in S$, uniquely. This is clear for $Sh_i(\{i\}, v)$. From here we can then proceed inductively. Suppose that we have defined $Sh_i(S, v)$ for all $S \subset I$, $S \neq I$, $i \in S$. We show that there is one and only one way to define $Sh_i(I, v)$, $i \in I$. To this effect, note that (18.AA.2) allows us to express every $Sh_i(I, v)$ as a function of $Sh_1(I, v)$ and of already determined numbers:

$$Sh_i(I, v) = Sh_1(I, v) + Sh_i(I \setminus \{1\}, v) - Sh_1(I \setminus \{i\}, v) \quad \text{for all } i \neq 1.$$

Then to determine $Sh_1(I, v)$ use $\sum_{i \in I} Sh_i(I, v) = v(I)$. Specifically,

$$Sh_1(I, v) = \frac{1}{I} \left[v(I) - \sum_{i \neq 1} Sh_i(I \setminus \{1\}, v) + \sum_{i \neq 1} Sh_1(I \setminus \{i\}, v) \right].$$

Definition 18.AA.7: The *Shapley value* of a game (I, v) , denoted

$$Sh(I, v) = (Sh_1(I, v), \dots, Sh_I(I, v)),$$

is the single outcome consistent with expression (18.AA.2).

We can compute $Sh_i(I, v)$ in a direct and interesting manner as follows. For any $S \subset I$ and $i \notin S$ let $m(S, i) = v(S \cup \{i\}) - v(S)$ be the *marginal contribution* of i to coalition S .³¹ For any ordering π of the players in I (technically, π is a one-to-one function from I to I) denote by $S(\pi, i) \subset I$ the set of players that come before i in the ordering π [technically, $S(\pi, i) = \{h: \pi(h) < \pi(i)\}$]. Note that, for any given ordering π , if we consider the marginal contributions of every player i to the set of the predecessors of i in the ordering π , then the sum of these marginal contributions must exactly exhaust $v(I)$; that is, $\sum_{i \in I} m(S(\pi, i), i) = v(I)$. It then turns out that $Sh_i(I, v)$ is the *average marginal contribution* of i to the set of her predecessors, where the average is taken over all orderings (held to be equally likely). Since the total number of orderings is $I!$ this gives

$$Sh_i(I, v) = \frac{1}{I!} \sum_{\pi} m(S(\pi, i), i). \quad (18.AA.3)$$

where the sum is taken over all orderings π of the players in I .

Example 18.AA.7: Glove Market. Consider the three-player game defined by

$$\begin{aligned} v(\{1, 2, 3\}) &= 1, \\ v(\{1, 3\}) &= v(\{2, 3\}) = 1, \quad v(\{1, 2\}) = 0, \\ v(\{1\}) &= v(\{2\}) = v(\{3\}) = 0. \end{aligned}$$

If the utility of a matched pair of gloves is 1, while an unmatched pair is worth nothing, then this game could arise from a situation in which players 1 and 2 own

31. Whenever we compute marginal contributions we follow the convention $v(\emptyset) = 0$. Therefore, $m(S, i) = v(\{i\})$ whenever $S = \emptyset$.

one right-hand glove each, while player 3 owns a left-hand glove. Let us compute $Sh_3(I, v)$. There are six possible orderings of the players:

$$\{1, 2, 3\}, \{1, 3, 2\}, \{2, 1, 3\}, \{2, 3, 1\}, \{3, 1, 2\}, \text{ and } \{3, 2, 1\}.$$

The marginal contribution of player 3 to its predecessors in each of these orderings is, respectively: 1, 1, 1, 1, 0, and 0. The average of these numbers is $\frac{2}{3}$; hence $Sh_3(I, v) = \frac{2}{3}$. Similarly, we get $Sh_1(I, v) = Sh_2(I, v) = \frac{1}{6}$. Note that these numbers satisfy (18.AA.2). For example,

$$Sh_3(I, v) - Sh_3(I \setminus \{1\}, v) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} - 0 = Sh_1(I, v) - Sh_1(I \setminus \{3\}, v). \quad \blacksquare$$

We can give a more explicit formula than (18.AA.3) for $Sh_i(I, v)$. The probability that in a random ordering a given coalition $T \subset I$, $i \in T$, arises as the union of i and its predecessors equals the probability that i is in the T th place,³² which is simply $1/I$, multiplied by the probability that $T \setminus \{i\}$ arises when we randomly select $\#T - 1$ members from the population $I \setminus \{i\}$, which is $(I - \#T)! (\#T - 1)! / (I - 1)!$. Hence, we can rewrite (18.AA.3) as

$$Sh_i(I, v) = \sum_{T \subset I, i \in T} [(I - \#T)! (\#T - 1)! / I!] (v(T) - v(T \setminus \{i\})). \quad (18.AA.4)$$

In Exercise 18.AA.4 you are asked to verify that if we were to define the Shapley value by (18.AA.4), or (18.AA.3), then equations (18.AA.2) would be satisfied; this means that, indeed, (18.AA.3) or (18.AA.4) provide correct formulas for the Shapley value.

We now put on record, rather informally, some of the basic properties of the Shapley value.

(a) *Efficiency*. $\sum_i Sh_i(I, v) = v(I)$; that is, no utility is wasted.

(b) *Symmetry*. If the games (I, v) and (I, v') are identical, except that the roles of players i and h are permuted,³³ then $Sh_i(I, v) = Sh_h(I, v')$. In words: The Shapley values do not depend on how we label players; only their position in the game, as summarized by the characteristic function, matters.

(c) *Linearity*. Note from (18.AA.3) or (18.AA.4) that the Shapley values depend linearly on the data, that is, on the coefficients $v(S)$ defining the game.

(d) *Dummy axiom*. Suppose that a player i contributes nothing to the game; that is, $v(S \cup \{i\}) - v(S) = 0$ for all $S \subset I$. Then $Sh_i(I, v) = 0$. This important property follows directly from (18.AA.3): The marginal contribution of player i to any coalition is null; hence its average is also null.

These four properties fully characterize the Shapley value. Although the proof of this fact is not difficult, we shall not give it here. See Exercises 18.AA.5 and 18.AA.6 for the discussion of some examples.

Given a game, the Shapley value assigns to it a single outcome. In contrast, the core solution assigns a set. We point out that the Shapley value need not belong to the

32. The symbol $\#T$ denotes the number of players in a coalition T .

33. Precisely, $v(S) = v'(S)$ whenever $i \in S$ and $h \in S$, $v(S) = v'(S)$ whenever $i \notin S$ and $h \notin S$, $v(S) = v'((S \setminus \{i\}) \cup \{h\})$ whenever $i \in S$ and $h \notin S$, and $v(S) = v'((S \setminus \{h\}) \cup \{i\})$ whenever $h \in S$ and $i \notin S$.

core. In a sense, we already know this because the Shapley value is defined for all games and there are games for which the core is empty. But the phenomenon can also occur if the core is nonempty. To see this, let us reexamine the glove market of Example 18.AA.7.

Example 18.AA.7 continued: In the glove market example a core utility outcome is $(0, 0, 1)$. Moreover, this is the *only* outcome in the core. Indeed, if (u_1, u_2, u_3) with $\sum_i u_i = 1$ has, say, $u_1 > 0$, then the coalition $\{2, 3\}$ can block by means of $(0, u_2 + \frac{1}{2}u_1, u_3 + \frac{1}{2}u_1)$. In effect, at the core the two owners of right-hand gloves undercut each other until they charge a price of zero. In contrast, the Shapley value, while heavily skewed towards player 3, nonetheless leaves something to the other two players ($\frac{1}{6}$ to each of them). ■

There is an important class of games for which the Shapley value belongs to the core. They are games characterized by the presence of a type of pronounced increasing returns to scale.

Definition 18.AA.8: A game (I, v) is *convex* if for every i the marginal contribution of i is larger to larger coalitions. Precisely, if $S \subset T$ and $i \in I \setminus T$, then

$$v(S \cup \{i\}) - v(S) \leq v(T \cup \{i\}) - v(T).$$

Example 18.AA.8: Complementary Inputs. Let $f(z_1, \dots, z_N)$ be a production function displaying increasing marginal productivities with respect to all inputs; that is, $\partial^2 f(z)/\partial z_h \partial z_k \geq 0$ for all z and h, k . Suppose that every player $i \in I$ is endowed with a vector of inputs $\omega_i \in \mathbb{R}_+^n$. Then we can define a TU-game by $v(S) = f(\sum_{i \in S} \omega_i)$. In Exercise 18.AA.8 you are asked to show that this game is convex. A warning on terminology: if $N = 1$ the previous condition simply says that $f(\cdot)$ is convex and, thus, the convexity of $f(\cdot)$ suffices for the convexity of the game; but for $N > 1$ the condition $\partial^2 f(z)/\partial z_h \partial z_k \geq 0$ for all z and h, k , is neither necessary nor sufficient for the convexity of $f(\cdot)$. In fact, the convexity of $f(\cdot)$ is far from sufficient for the convexity of the game (see Exercise 18.AA.8). ■

We can then show the result in Proposition 18.AA.1.

Proposition 18.AA.1: If a game (I, v) is convex then its Shapley value utility outcome $Sh(I, v) = (Sh_1(I, v), \dots, Sh_I(I, v))$ belongs to the core (in particular, the core is nonempty).

Proof: It is enough to show that if $i \in S \subset T$ then $Sh_i(S, v) \leq Sh_i(T, v)$. Indeed, for any $S \subset I$ this implies that $v(S) = \sum_{i \in S} Sh_i(S, v) \leq \sum_{i \in S} Sh_i(I, v)$ and therefore the coalition S cannot block.

To prove the claimed property it suffices to consider $i \in S$ and $T = S \cup \{h\}$. Given an ordering π of S denote by $m(\pi, i)$ the marginal contribution of i to its predecessors in S and according to the ordering π by $m'(\pi, i)$ the average marginal contribution of i to its predecessors in T when the average is taken over the $\#T$ orderings of T differing from the given ordering π of S only by the placement of h . Then

$$Sh_i(S, v) = \frac{1}{\#S!} \sum_{\pi} m(\pi, i) \quad \text{and} \quad Sh_i(T, v) = \frac{1}{\#S!} \sum_{\pi} m'(\pi, i).$$

Note that for every ordering π of S we must have $m'(\pi, i) \geq m(\pi, i)$: If we place h after i then

the marginal contribution of i to its predecessors in T is still $m(\pi, i)$; if we place h before i then, by the convexity condition, this marginal contribution is at least $m(\pi, i)$. We conclude that $Sh_i(T, v) \geq Sh_i(S, v)$, as we wanted to show. ■

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EXERCISES

18.B.1^A Show that Walrasian allocations are in the core for the model with a constant returns technology described in Section 18.B.

18.B.2^B Exhibit an example of a nonequal-treatment core allocation in a three-consumer exchange economy with continuous, strictly convex, strongly monotone preferences. Can an example be given with only two consumers?

18.B.3^A Give a direct proof (i.e., not using properties of the core) that a Walrasian allocation of an economy with continuous, strictly convex preferences has the equal-treatment property.

18.B.4^A Use Taylor's formula to complete the proof of Proposition 18.B.3.

18.B.5^B Consider an economy composed of $2I + 1$ consumers. Of these, I each own one right shoe and $I + 1$ each own a left shoe. Shoes are indivisible. Everyone has the same utility function, which is $\text{Min} \{R, L\}$, where R and L are, respectively, the quantities of right and left shoes consumed.

(a) Show that any allocation of shoes that is matched (i.e., every individual consumes the same number of shoes of each kind) is a Pareto optimum, and conversely.

(b) Which Pareto optima are in the core of this economy? (This time, in the definition of the core allow for weak dominance in blocking.)

(c) Let p_R and p_L be the respective prices of the two kinds of shoes. Find the Walrasian equilibria of this economy.

(d) Comment on the relationship between the core and the Walrasian equilibria in this economy.

18.C.1^C Establish the properties of effective budget sets claimed in the discussion of Example 18.C.3. You can restrict yourself to the case $L = 2$.

18.D.1^B Consider an Edgeworth box with continuous, strictly convex and monotone preferences. Show that every feasible allocation where both consumers are at least as well off as at their initial endowments is self-selective.

18.E.1^B In text.

18.E.2^A Use the envelope theorem (see Section M.L of the Mathematical Appendix) to derive expression (18.E.5).

18.E.3^B By considering an example with L -shaped preferences for two non-numeraire goods (hence, the utility function cannot be differentiable), argue that it is possible that at a Walrasian allocation with a continuum of traders every trader gets less than her marginal contribution.

18.AA.1^B A collection of coalitions $S_1, \dots, S_N \subset I$ is a *generalized partition* if we can assign a weight $\delta_n \in [0, 1]$ to every $1 \leq n \leq N$ such that, for every player $i \in I$, we have $\sum_{\{n: i \in S_n\}} \delta_n = 1$. Exhibit examples of generalized partitions, with the corresponding weights.

We say that a TU-game (I, v) is *balanced* if for every generalized partition we have $\sum_n \delta_n v(S_n) \leq v(I)$, where δ_n are the corresponding partition weights. Show that the game has a nonempty core if and only if it is balanced. [Hint: Appeal to the duality theorem of linear programming (see Section M.M of the Mathematical Appendix).]

18.AA.2^A In text.

18.AA.3^A Show that the proportional allocation of Example 18.AA.6 is the only allocation in the core if average product is constant.

18.AA.4^C Show that if the Shapley value is defined by formula (18.AA.4)—or, equivalently, by (18.AA.3)—then the preservation of differences expression (18.AA.2) is satisfied.

18.AA.5^B We say that a game (I, v) is a *unanimity* game if there is a nonempty $S \subset I$ such that $v(T) = v(S)$ if $S \subset T$ and $v(T) = 0$ otherwise. Show then that under the efficiency, symmetry, and dummy axioms we are led to distribute $v(S)$ equally across the members of S .

18.AA.6^B Show that any TU-game (I, v) can be expressed as a linear combination of unanimity games. Then use the Exercise 18.AA.5 and the linearity axiom to show that there is a unique solution satisfying the efficiency, symmetry, dummy, and linearity axioms. Connect your discussion with the Shapley value.

18.AA.7^C Show that the production game described in Example 18.AA.8 is convex.

18.AA.8^B In the context of the production example of Example 18.AA.8, give an example of a two-input production function that is convex (as a function) but for which, nonetheless, the core is empty (thus, the induced game cannot be convex).

18.AA.9^B Consider the game with four players defined by $v(\{i\}) = 0$, $v(\{12\}) = v(\{34\}) = 0$, $v(\{13\}) = v(\{14\}) = v(\{23\}) = v(\{24\}) = 1$, $v(\{ijk\}) = 1$ for all three-player coalitions $\{ijk\}$, and $v(\{1234\}) = 2$.

(a) Show that this is the game that you would get from the utility production technology $\text{Min}\{z_1, z_2\}$, where z_1 and z_2 are the amounts of two factors, if the factor endowments of the four consumers are $\omega_1 = \omega_2 = (1, 0)$ and $\omega_3 = \omega_4 = (0, 1)$.

(b) Show that the core of this game contains all points of the form $(\alpha, \alpha, 1 - \alpha, 1 - \alpha)$ for $\alpha \in [0, 1]$.

(c) Show that if $v(\{134\})$ is increased to 2, holding all other coalition values constant, there is then only one point in the core. Compare the welfare of player 1 at this point to what she would get at all the points in the core before the increase in $v(\{134\})$.

(d) Compute the Shapley value of the game [before the modification in (c)] without using the brute-force enumeration technique. [Hint: Use symmetry considerations and other axiomatically based simplifications to go part of the way to the answer.]

(e) How does the Shapley value change under the modification of part (c)? Discuss the difference between the changes in the Shapley value and in the core.

18.AA.10^B Consider a firm constituted by two divisions. The firm must provide overhead in the form of space, (x_1, x_2) , to each of them. The cost of *aggregate* amounts of space is given by $C(x_1 + x_2) = (x_1 + x_2)^\gamma$, $0 < \gamma < 1$.

(a) Suppose that, whatever the usage of space (x_1, x_2) , the total cost must be *exactly* allocated between the two divisions. Propose a cost allocation system based on the Shapley value to accomplish this.

(b) Compute the marginal cost imposed on each of the two divisions [according to the cost allocation system identified in (a)] whenever a division increases its usage of space.

(c) Suppose now that the profits accruing to the two divisions are $\alpha_1 x_1$ and $\alpha_2 x_2$, respectively (we assume that $\alpha_1 > 0$ and $\alpha_2 > 0$), and that each division uses space to the point where marginal profits equal own marginal costs [as determined in (b)]. Will this lead to an efficient (that is, profit-maximizing) choice of overhead?

(d) Is there any distribution rule $\psi_1(x_1, x_2)$, $\psi_2(x_1, x_2)$, with $\psi_1(x_1, x_2) + \psi_2(x_1, x_2) = C(x_1 + x_2)$ for all (x_1, x_2) , that leads to efficient decentralized choice [in the sense of (c)] for all α_1, α_2 ? [Hint: Consider the externality imposed by each division on the other.]

General Equilibrium Under Uncertainty

19.A Introduction

In this chapter, we apply the general equilibrium framework developed in Chapters 15 to 18 to economic situations involving the exchange and allocation of resources under conditions of uncertainty. In a sense, this chapter offers the equilibrium counterpart of the decision theory presented in Chapter 6 (and which we recommend you review at this point).

We begin, in Section 19.B, by formalizing uncertainty by means of *states of the world* and then introducing the key idea of a *contingent commodity*: a commodity whose delivery is conditional on the realized state of the world. In Section 19.C we use these tools to define the concept of an *Arrow–Debreu equilibrium*. This is simply a Walrasian equilibrium in which contingent commodities are traded. It follows from the general theory of Chapter 16 that an Arrow–Debreu equilibrium results in a Pareto optimal allocation of risk.

In Section 19.D, we provide an important reinterpretation of the concept of Arrow–Debreu equilibrium. We show that, under the assumptions of *self-fulfilling, or rational, expectations*, Arrow–Debreu equilibria can be implemented by combining trade in a certain restricted set of contingent commodities with *spot trade* that occurs *after* the resolution of uncertainty. This results in a significant reduction in the number of *ex ante* (i.e., before uncertainty) markets that must operate.

In Section 19.E, we generalize our analysis. Instead of trading contingent commodities prior to the resolution of uncertainty, agents now trade *assets*; and instead of an Arrow–Debreu equilibrium we have the notion of a *Radner equilibrium*. We also discuss here the important notion of *arbitrage* among assets. The material of this section lies at the foundations of a very rich body of finance theory [good introductions are Duffie (1992) and Huang and Litzenberger (1988)].

In Section 19.F, we briefly illustrate some of the welfare difficulties raised by the possibility of *incomplete markets*, that is, by the possibility of there being too few asset markets to guarantee a fully Pareto optimal allocation of risk.

Section 19.G is devoted to the issue of the objectives of the firm under conditions of uncertainty. In particular, it gives sufficient conditions for shareholders to agree unanimously on the objective of *market value maximization*.

Section 19.H takes a close look at the informational requirements of the theory developed in this chapter. We see that the theory applies well to situations of *symmetric* information across consumers (reviewed in Section 19.H); but its applicability is more problematic in situations of *asymmetric* information. This provides a further argument for the techniques developed in Chapters 13 and 14 for the study of asymmetric information problems.

For additional material and references on the topic of this chapter, see the textbooks of Huang and Litzenberger (1988) and Duffie (1992) already mentioned, or, at a more advanced level, Radner (1982) and Magill and Shafer (1991).

19.B A Market Economy with Contingent Commodities: Description

As in our previous chapters, we contemplate an environment with L physical commodities, I consumers, and J firms. The new element is that technologies, endowments, and preferences are now *uncertain*.

Throughout this chapter, we represent uncertainty by assuming that technologies, endowments, and preferences depend on the *state of the world*. The concept of state of the world was already introduced in Section 6.E. A state of the world is to be understood as a complete description of a possible outcome of uncertainty, the description being sufficiently fine for any two distinct states of the world to be mutually exclusive. We assume that an exhaustive set S of states of the world is given to us. For simplicity we take S to be a finite set with (abusing notation slightly) S elements. A typical element is denoted $s = 1, \dots, S$.

We state in Definition 19.B.1 the key concepts of a *(state-)contingent commodity* and a *(state-)contingent commodity vector*. Using these concepts we shall then be able to express the dependence of technologies, endowments, and preferences on the realized states of the world.

Definition 19.B.1: For every physical commodity $\ell = 1, \dots, L$ and state $s = 1, \dots, S$, a unit of *(state-)contingent commodity* ℓs is a title to receive a unit of the physical good ℓ if and only if s occurs. Accordingly, a *(state-)contingent commodity vector* is specified by

$$x = (x_{11}, \dots, x_{L1}, \dots, x_{1S}, \dots, x_{LS}) \in \mathbb{R}^{LS},$$

and is understood as an entitlement to receive the commodity vector $(x_{1s}, \dots, x_{Ls})$ if state s occurs.¹

We can also view a contingent commodity vector as a collection of L *random variables*, the ℓ th random variable being $(x_{\ell 1}, \dots, x_{\ell S})$.

With the help of the concept of contingent commodity vectors, we can now describe how the characteristics of economic agents depend on the state of the world. To begin, we let the endowments of consumer $i = 1, \dots, I$ be a contingent commodity vector:

$$\omega_i = (\omega_{11i}, \dots, \omega_{L1i}, \dots, \omega_{1Si}, \dots, \omega_{LSi}) \in \mathbb{R}^{LS}.$$

1. As usual, a negative entry is understood as an obligation to deliver.